CHAPTER 6

NUCLEAR REACTOR THEORY

In a critical reactor, as explained in Section 4.1, there is a balance between the number of neutrons produced in fission and the number lost, either by absorption in the reactor or by leaking from its surface. One of the central problems in the design of a reactor is the calculation of the size and composition of the system required to maintain this balance. This problem is the subject of the present chapter.

Calculations of the conditions necessary for criticality are normally carried out using the group diffusion method introduced at the end of the last chapter. It is reasonable to begin the present discussion with a one-group calculation. While this type of calculation is most appropriate for fast reactors, it will be shown later in the chapter that the one-group method can also be used in a modified form for computations of some thermal reactors.

6.1 ONE-GROUP REACTOR EQUATION

Consider a critical fast reactor containing a homogeneous mixture of fuel and coolant. It will be assumed that the reactor consists of only one region, and has neither a blanket nor a reflector. Such a system is said to be a *bare reactor*.

This reactor is described in a one-group calculation by the one-group diffusion equation (see Eq. 5.52)

$$D\nabla^2\phi - \Sigma_a\phi = -s. \tag{6.1}$$

Here, ϕ is the one-group flux, D and Σ_a are the one-group diffusion coefficient and macroscopic absorption cross section for the fuel-coolant mixture, and s is the source density, that is, the number of neutrons emitted per cm^3/sec. Nominal one-group cross sections for a fast reactor are given in Table 6.1.

TABLE 6.1
Nominal one-group constants for a fast reactor*

Element or isotope	σ_γ	σ_f	σ_a	σ_{tr}	ν	η
Na	0.0008	0	0.0008	3.3	—	—
Al	0.002	0	0.002	3.1	—	—
Fe	0.006	0	0.006	2.7	—	—
^{235}U	0.25	1.4	1.65	6.8	2.6	2.2
^{238}U	0.16	0.095	0.255	6.9	2.6	0.97
^{239}Pu	0.26	1.85	2.11	6.8	2.98	2.61

*From *Reactor Physics Constants*, 2nd ed., Argonne National Laboratory report ANL-5800, 1963.

In a critical reactor, the source neutrons are emitted in fission. To determine s, let Σ_{aF} be the one-group absorption cross section of the fuel, and let η be the average number of fission neutrons emitted per neutron absorbed in the fuel.* The source term is then given by

$$s = \eta \Sigma_{aF} \phi. \tag{6.2}$$

This may also be written as

$$s = \eta \frac{\Sigma_{aF}}{\Sigma_a} \Sigma_a \phi = \eta f \Sigma_a \phi, \tag{6.3}$$

where

$$f = \frac{\Sigma_{aF}}{\Sigma_a} \tag{6.4}$$

is called the *fuel utilization*. Since Σ_a is the cross section for the mixture of fuel and coolant, while Σ_{aF} is the cross section of the fuel only, it follows that f is equal to the fraction of the neutrons absorbed in the reactor that are absorbed by the fuel.

The source term in Eq. (6.3) can also be written in terms of the multiplication factor, which was defined in Chapter 4. For this purpose, consider an infinite reactor having the same composition as the bare reactor under discussion. With such a reactor, there can be no escape of neutrons, as there is from the surface of a bare reactor. All neutrons eventually are absorbed, either in the fuel or in the coolant. Furthermore, the neutron flux must be a constant, independent of position. It follows then that $\Sigma_a \phi$ neutrons are absorbed per cm^3/sec everywhere in the system. Of these neutrons, the number $f \Sigma_a \phi$ are absorbed in fuel and release $\eta f \Sigma_a \phi$ fission

*If the fuel consists of more than one species of fissile nuclide or a mixture of fissile and fissionable nuclides, then η should be computed as in Section 3.7 using Eq. (3.49).

neutrons, all of which, sooner or later, are absorbed in the reactor. Thus the absorption of $\Sigma_a\phi$ neutrons in one generation leads to the absorption of $\eta f\Sigma_a\phi$ in the next. Now according to the discussion in Section 4.1, the multiplication factor is defined as the number of fissions—or any other event at some point in a chain reaction—in one generation divided by the number in the preceding generation. It follows, therefore, that

$$k_\infty = \frac{\eta f \Sigma_a \phi}{\Sigma_a \phi} = \eta f, \tag{6.5}$$

where the subscript on k_∞ signifies that this result is valid only for the infinite reactor.

Example 6.1 Calculate f and k_∞ for a mixture of ^{235}U and sodium in which the uranium is present to 1 $^{w}/_{o}$.

Solution. From Eq. (6.4) f is

$$f = \frac{\Sigma_{aF}}{\Sigma_a} = \frac{\Sigma_{aF}}{\Sigma_{aF} + \Sigma_{aS}},$$

where Σ_{aF} and Σ_{aS} are the macroscopic absorption cross sections of the uranium and sodium, respectively. Dividing numerator and denominator by Σ_{aF} gives

$$f = \frac{1}{1 + \Sigma_{aS}/\Sigma_{aF}} = \frac{1}{1 + N_S\sigma_{aS}/N_F\sigma_{aF}},$$

where N_F and N_S are the atom densities of the uranium and sodium. Next let ρ_F and ρ_S be the number of grams of uranium and sodium per cm^3 in the mixture. Then

$$\frac{N_S}{N_F} = \frac{\rho_S}{\rho_F}\frac{M_F}{M_S},$$

where M_F and M_S are the gram atomic weights of uranium and sodium. Since 1 $^{w}/_{o}$ of the mixture is fuel, this means that

$$\frac{\rho_F}{\rho_F + \rho_S} = 0.01$$

or

$$\frac{\rho_S}{\rho_F} = 99.$$

Using the values of σ_a given in Table 6.1, the value of f is given by

$$f^{-1} = 1 + 99 \times \frac{235}{23} \times \frac{0.0008}{1.65} = 1.49,$$

and

$$f = 0.671. \ [Ans.]$$

The value of k_∞ is

$$k_\infty = \eta f = 2.2 \times 0.671 = 1.48.$$

Since this is greater than unity, an infinite reactor with this composition would be supercritical. [*Ans.*]

Since η and f are constants which depend only on the material properties of the reactor, the value of k_∞ is the same for a bare reactor as for an infinite reactor of the same composition. The source term in Eq. (6.3) can therefore be written as

$$s = k_\infty \Sigma_a \phi. \tag{6.6}$$

Introducing Eq. (6.6) into the one-group diffusion equation, Eq. (6.1), gives

$$D\nabla^2\phi - \Sigma_a\phi = -k_\infty\Sigma_a\phi$$

or

$$D\nabla^2\phi + (k_\infty - 1)\Sigma_a\phi = 0. \tag{6.7}$$

Dividing by D yields

$$\nabla^2\phi + \frac{k_\infty - 1}{L^2}\phi = 0, \tag{6.8}$$

where

$$L^2 = \frac{D}{\Sigma_a} \tag{6.9}$$

is the one-group diffusion area. Finally, it is convenient to introduce the parameter B^2, defined as

$$B^2 = \frac{k_\infty - 1}{L^2}. \tag{6.10}$$

Substituting this expression into Eq. (6.8) gives

$$\nabla^2\phi + B^2\phi = 0. \tag{6.11}$$

This important result is known as the *one-group reactor equation*. It will be shown in the next sections that this equation, together with the usual boundary conditions on ϕ, not only determines the shape of the flux in the reactor, but also leads to a condition which must be satisfied in order that the reactor be critical.

6.2 THE SLAB REACTOR

As the first example of a bare reactor, consider a system consisting of an infinite bare slab of thickness a as shown in Fig. 6.1. The reactor equation in this case is

$$\frac{d^2\phi}{dx^2} + B^2\phi = 0, \tag{6.12}$$

where x is measured from the center of the slab.

In order to determine the flux within the reactor, Eq. (6.12) must be solved subject to the boundary condition that ϕ vanish at the extrapolated faces of the slab, that is, at $x = a/2 + d$ and at $x = -a/2 - d$. For simplicity, however, d will be taken to be zero—in most practical problems, d is much smaller than reactor dimensions. Then the boundary conditions become

$$\phi\left(\frac{a}{2}\right) = \phi\left(-\frac{a}{2}\right) = 0. \tag{6.13}$$

It may also be noted that because of the symmetry of the problem there can be no net flow of neutrons at the center of the slab. Since the neutron current density is proportional to the derivative of ϕ, this means that

$$\frac{d\phi}{dx} = 0 \tag{6.14}$$

at $x = 0$. The condition given by Eq. (6.14) is equivalent to requiring that ϕ be an even function, that is,

$$\phi(-x) = \phi(x), \tag{6.15}$$

and have a continuous derivative within the reactor. [In any problem where it is clear that the flux is a well-behaved function, Eq. (6.15) is often easier to apply than Eq. (6.14).]

In any case, the general solution to Eq. (6.12) is

$$\phi(x) = A\cos Bx + C\sin Bx, \tag{6.16}$$

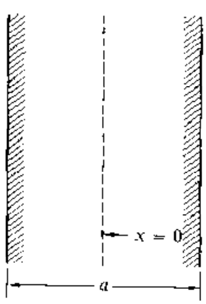

Fig. 6.1 The infinite slab reactor.

where A and C are constants to be determined. Placing the derivative of Eq. (6.16) equal to zero at $x = 0$ gives immediately $C = 0$, so that ϕ reduces to

$$\phi(x) = A \cos Bx.$$

Next, introducing the boundary condition given by Eq. (6.13) at $a/2$ or $-a/2$ (it makes no difference which, since the cosine is an even function) gives

$$\phi\left(\frac{a}{2}\right) = A \cos\left(\frac{Ba}{2}\right) = 0. \tag{6.17}$$

This equation can be satisfied either by taking $A = 0$, which leads to the trivial solution $\phi(x) = 0$, or by requiring that

$$\cos\left(\frac{Ba}{2}\right) = 0. \tag{6.18}$$

This, in turn, will be satisfied if B assumes any of the values B_n, where

$$B_n = \frac{n\pi}{a} \tag{6.19}$$

and n is an odd integer, as can readily be seen by direct substitution into Eq. (6.18).

The various constants B_n are known as *eigenvalues* and the corresponding functions $\cos B_n x$ are called *eigenfunctions*. It can be shown that if the reactor under consideration were not critical, the flux would be the sum of all such eigenfunctions, each multiplied by a function which depends on the time. However, if the reactor is critical, all of these functions except the first die out in time and the flux assumes the steady-state shape of the first eigenfunction or *fundamental*, namely,

$$\phi(x) = A \cos B_1 x = A \cos\left(\frac{\pi x}{a}\right). \tag{6.20}$$

This is the flux in a critical slab reactor.

The square of the lowest eigenvalue B_1^2 is called the *buckling* of the reactor. The origin of this term can be seen by solving the equation satisfied by the flux, namely,

$$\frac{d^2\phi}{dx^2} + B_1^2\phi = 0, \tag{6.21}$$

for B_1^2. The result is

$$B_1^2 = -\frac{1}{\phi}\frac{d^2\phi}{dx^2}.$$

The right-hand side of this expression is proportional to the curvature of the flux in the reactor, which, in turn, is a measure of the extent to which the flux curves or "buckles." Since in the slab reactor

$$B_1^2 = \left(\frac{\pi}{a}\right)^2, \tag{6.22}$$

the buckling decreases as a increases. In the limit as a becomes infinite, $B_1^2 = 0$, ϕ is a constant and has no "buckle."

It should be observed that the value of the constant A in Eq. (6.20), which determines the magnitude of ϕ, has not been established in the above analysis. Mathematically, this is because the reactor equation (Eq. 6.11 or 6.12) is homogeneous, and ϕ multiplied by any constant is still a solution to the equation. Physically the reason why the value of A has not been established is that the magnitude of the flux in a reactor is determined by the *power* at which the system is operating, and not by its material properties.

To find an expression for A, it is necessary to make a separate calculation of the reactor power. In particular, there are $\Sigma_f \phi(x)$ fissions per cm^3/sec at the point x, where Σ_f is the macroscopic fission cross section, and if the recoverable energy is E_R joules per fission* (with a recoverable energy of 200 MeV, $E_R = 3.2 \times 10^{-11}$ joules), then the total power per unit area of the slab, in watts/cm^2, is

$$P = E_R \Sigma_f \int_{-a/2}^{a/2} \phi(x)\, dx. \tag{6.23}$$

Inserting the expression for $\phi(x)$ from Eq. (6.20) and performing the integration gives

$$P = \frac{2aE_R\Sigma_f A}{\pi} \tag{6.24}$$

The final formula for the thermal flux in a slab reactor is then

$$\phi(x) = \frac{\pi P}{2aE_R\Sigma_f} \cos\left(\frac{\pi x}{a}\right). \tag{6.25}$$

6.3 OTHER REACTOR SHAPES

It is not possible, of course, to construct a reactor in the form of an infinite slab, and it is necessary, therefore, to generalize the results of the preceding section to reactors of more realistic shapes. This can easily be done for the following reactors: sphere, infinite cylinder, rectangular parallelepiped, and finite cylinder. These reactors, together with the coordinate systems used to describe them, are shown in Fig. 6.2. It should be noted that each of these reactors is bare; reflected reactors will be discussed in Section 6.7.

*In the author's book on reactor theory, E_R, when expressed in joules, was denoted by γ.

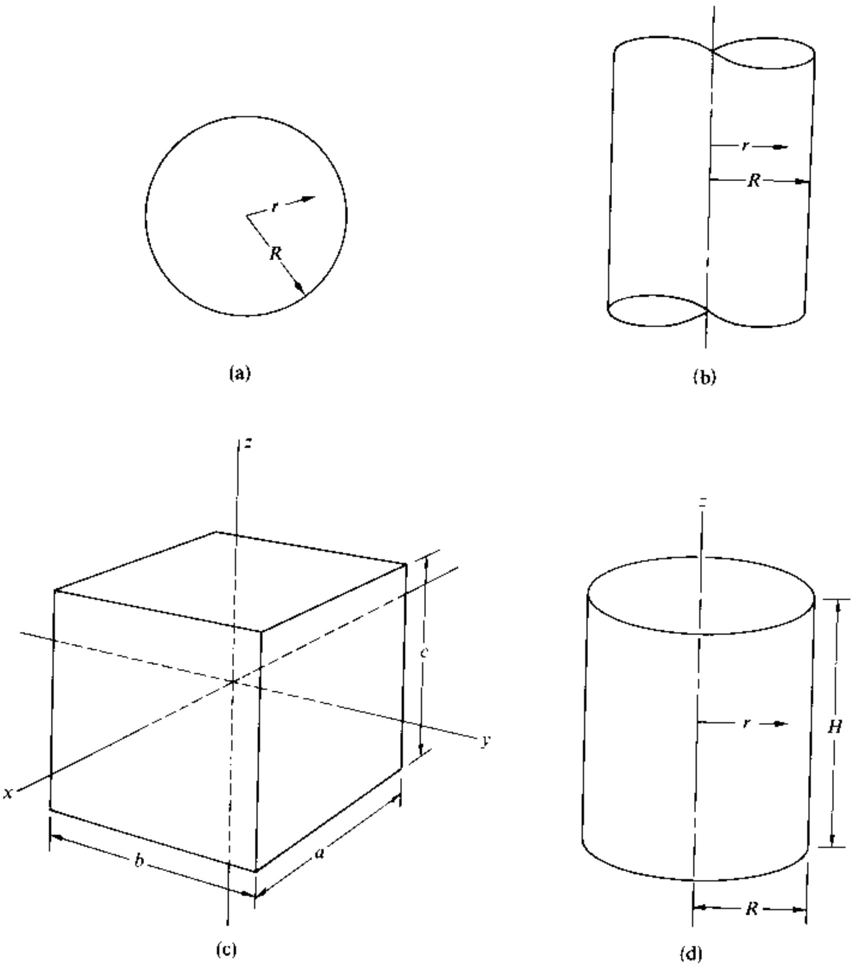

Fig. 6.2 (a) The spherical reactor; (b) the infinite cylindrical reactor; (c) the parallelepiped reactor; (d) the finite cylindrical reactor.

Sphere

Consider first a spherical reactor of radius R. The flux in this reactor is a function only of r, and the reactor equation is

$$\frac{1}{r^2}\frac{d}{dr}r^2\frac{d\phi}{dr} + B^2\phi = 0, \tag{6.26}$$

where use has been made of the Laplacian in spherical coordinates. The flux must satisfy the boundary condition $\phi(R) = 0$, the extrapolation distance being ignored, and remain finite throughout the interior of the reactor.

By substituting $\phi = w/r$ into Eq. (6.26) and solving the resulting equation for w as was done in Section 5.6, the general solution to Eq. (6.26) is easily found to be

$$\phi = A\frac{\sin Br}{r} + C\frac{\cos Br}{r},$$

where A and C are constants. The second term becomes infinite when r goes to zero, however, and C must be placed equal to zero. Thus ϕ becomes

$$\phi = A\frac{\sin Br}{r}. \tag{6.27}$$

The boundary condition $\phi(R) = 0$ can be satisfied by taking B to be any one of the eigenvalues

$$B_n = \frac{n\pi}{R},$$

where n is any integer. However, as explained in the preceding section, only the first eigenvalue is relevant for a critical reactor. Thus with $n = 1$, the buckling is

$$B_1^2 = \left(\frac{\pi}{R}\right)^2 \tag{6.28}$$

and the flux becomes

$$\phi = A\frac{\sin(\pi r/R)}{r}. \tag{6.29}$$

The constant A is again determined by the operating power of the reactor, namely

$$P = E_R\Sigma_f \int \phi(r)\, dV, \tag{6.30}$$

where dV is the differential volume element. In view of the geometry of the problem, dV is given by

$$dV = 4\pi r^2\, dr$$

and Eq. (6.30) becomes

$$P = 4\pi E_R\Sigma_f \int_0^R r^2\phi(r)\, dr.$$

Introducing the flux from Eq. (6.29) and carrying out the integration gives

$$P = 4E_R\Sigma_f AR^2.$$

The flux in the sphere may therefore be written as

$$\phi = \frac{P}{4E_R\Sigma_f R^2}\frac{\sin(\pi r/R)}{r}. \tag{6.31}$$

Infinite Cylinder

Next consider an infinite cylindrical reactor of radius R. In this reactor the flux depends only on the distance r from the axis. With the Laplacian appropriate to cylindrical coordinates described in Appendix III, the reactor equation becomes

$$\frac{1}{r}\frac{d}{dr}r\frac{d\phi}{dr} + B^2\phi = 0,$$

or, when the differentiation in the first term is carried out,

$$\frac{d^2\phi}{dr^2} + \frac{1}{r}\frac{d\phi}{dr} + B^2\phi = 0. \tag{6.32}$$

Besides satisfying this equation, ϕ must also satisfy the usual boundary conditions including $\phi(R) = 0$.

Equation (6.32) is a special case of *Bessel's equation,*

$$\frac{d^2\phi}{dr^2} + \frac{1}{r}\frac{d\phi}{dr} + \left(B^2 - \frac{m^2}{r^2}\right)\phi = 0, \tag{6.33}$$

in which m is a constant. Since Eq. (6.33) is a second-order differential equation it has two independent solutions. These are denoted as $J_m(Br)$ and $Y_m(Br)$ and are called *ordinary Bessel functions of the first and second kind,* respectively. These functions appear in many engineering and physics problems and they have been widely tabulated.*

Comparing Eq. (6.32) with Eq. (6.33) shows that in the present problem m is equal to zero. The general solution to the reactor equation can therefore be written as

$$\phi = AJ_0(Br) + CY_0(Br),$$

where A and C are again constants. The functions $J_0(x)$ and $Y_0(x)$ are plotted in Fig. 6.3. It will be observed that $Y_0(x)$ is infinite at $x = 0$, while $J_0(0) = 1$. Therefore, since ϕ must remain finite within the reactor, C must be taken to be zero. Thus ϕ reduces to

$$\phi = AJ_0(Br). \tag{6.34}$$

The boundary condition $\phi(R) = 0$ now becomes

$$\phi(R) = AJ_0(Br) = 0. \tag{6.35}$$

*A short table of Bessel functions is given in Appendix V.

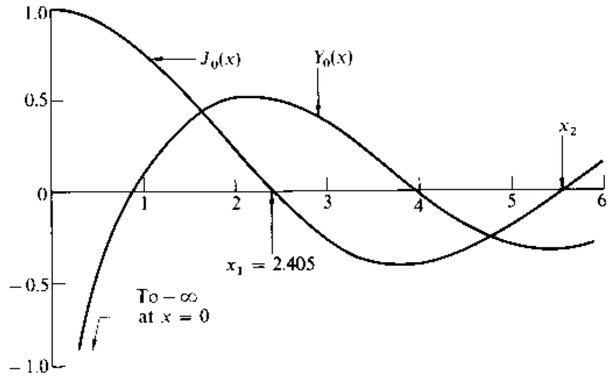

Fig. 6.3 The Bessel functions $J_0(x)$ and $Y_0(x)$.

As shown in Fig. 6.3, the function $J_0(x)$ is equal to zero at a number of values of x, labeled $x_1, x_2, \ldots$, so that $J_0(x_n) = 0$. This means that Eq. (6.35) will be satisfied provided B is any one of the values

$$B_n = \frac{x_n}{R},$$

which are the eigenvalues of the problem. However, since in a critical reactor only the lowest eigenvalue is important, it follows that the buckling is

$$B_1^2 = \left(\frac{x_1}{R}\right)^2 = \left(\frac{2.405}{R}\right)^2 \tag{6.36}$$

The one-group flux is then

$$\phi = AJ_0\left(\frac{2.405r}{R}\right) \tag{6.37}$$

The constant A is again determined by Eq. (6.30), where for the infinite cylinder $dV = 2\pi r dr$. Thus the power per unit length of the cylinder is

$$\begin{aligned} P &= 2\pi E_R \Sigma_f \int_0^R \phi(r) r\, dr \\ &= 2\pi E_R \Sigma_f A \int_0^R J_0\left(\frac{2.405r}{R}\right) r\, dr. \end{aligned}$$

This integral can be evaluated using the formula

$$\int_0^x J_0(x')x'\, dx' = xJ_1(x),$$

which gives

$$P = 2\pi E_R \Sigma_f R^2 A J_1(2.405)/2.405 = 1.35 E_R \Sigma_f R^2 A.$$

TABLE 6.2
Bucklings and fluxes for critical bare reactors

Geometry	Dimensions	Buckling	Flux	A	ϕ_{max}/ϕ_{av}
Infinite slab	Thickness a	$\left(\frac{\pi}{a}\right)^2$	$A\cos\left(\frac{\pi x}{a}\right)$	$1.57P/aE_R\Sigma_f$	1.57
Rectangular parallelepiped	$a \times b \times c$	$\left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2 + \left(\frac{\pi}{c}\right)^2$	$A\cos\left(\frac{\pi x}{a}\right)\cos\left(\frac{\pi y}{b}\right)\cos\left(\frac{\pi z}{c}\right)$	$3.87P/VE_R\Sigma_f$	3.88
Infinite cylinder	Radius R	$\left(\frac{2.405}{R}\right)^2$	$AJ_0\left(\frac{2.405r}{R}\right)$	$0.738P/R^2E_R\Sigma_f$	2.32
Finite cylinder	Radius R Height H	$\left(\frac{2.405}{R}\right)^2 + \left(\frac{\pi}{H}\right)^2$	$AJ_0\left(\frac{2.405r}{R}\right)\cos\left(\frac{\pi z}{H}\right)$	$3.63P/VE_R\Sigma_f$	3.64
Sphere	Radius R	$\left(\frac{\pi}{R}\right)^2$	$A\frac{1}{r}\sin\left(\frac{\pi r}{R}\right)$	$P/4R^2E_R\Sigma_f$	3.29

The final expression for the flux is then

$$\phi = \frac{0.738P}{E_R \Sigma_f R^2} J_0\left(\frac{2.405r}{R}\right). \tag{6.38}$$

Other Geometries

The derivations of the flux distributions in the parallelepiped and finite cylinder reactors are somewhat lengthy and will not be given here. Values of the bucklings and fluxes for these reactors are given in Table 6.2, which also summarizes the results obtained earlier for the slab, sphere, and infinite cylinder.

Maximum to Average Flux and Power

The maximum value of the flux, ϕ_{max}, in a uniform bare reactor is always found at the center of the reactor. Since the power density (e.g., watts per cm^3) is proportional to the flux, this means that the power density is also highest at the center. It is of some interest to compute the ratio of the maximum flux to its average value throughout the reactor. This ratio, which will be denoted by Ω, is a measure of the overall variation of the flux within a reactor and is an indication of the extent to which the power density at the center of the system exceeds the average power density.

Consider the case of a bare spherical reactor. The value of ϕ_{max} is obtained by taking the limit of Eq. (6.31) as r goes to zero. Thus

$$\phi_{max} = \frac{P}{4E_R \Sigma_f R^2} \lim_{r \to 0} \frac{\sin(\pi r/R)}{r} = \frac{\pi P}{4E_R \Sigma_f R^3}. \tag{6.39}$$

The average value of ϕ is given by

$$\phi_{av} = \frac{1}{V} \int \phi \, dV, \tag{6.40}$$

where the integral is carried out over the volume of the sphere. However, this integral is proportional to the reactor power, that is,

$$P = E_R \Sigma_f \int \phi \, dV,$$

and so,

$$\phi_{av} = \frac{P}{E_R \Sigma_f V}, \tag{6.41}$$

a result valid for all geometries. Dividing Eq. (6.39) by Eq. (6.41) gives for the sphere

$$\Omega = \frac{\phi_{max}}{\phi_{av}} = \frac{\pi^2}{3} = 3.29. \tag{6.42}$$

Values of Ω for other geometries are given in Table 6.2.

For reasons to be discussed in the next two chapters, it is important that the flux distribution in an actual reactor be as uniform, that is, as *flat* as possible. The values of Ω in Table 6.2 would be quite unacceptable for a real reactor. However, it must be borne in mind that these values were derived for bare reactors. In such systems, the flux vanishes at the extrapolated surface of the core and ϕ therefore undergoes large changes from the reactor center to its surface. With reflected reactors, on the other hand, as will be shown in Section 6.6, the flux does not fall to low values at the core–reflector interface and the flux in these reactors (and all power reactors are reflected) is considerably flatter. The flux is made still flatter in many modern reactors by non-uniformly distributing the fuel in the core (see Section 7.5).

Example 6.2 A bare spherical reactor of radius 50 cm operates at a power level of 100 megawatts $= 10^8$ joules/sec. If $\Sigma_f = 0.0047\ \text{cm}^{-1}$, what are the maximum and average values of the flux in the reactor?

Solution. The maximum value of ϕ can be found directly from Eq. (6.39). Introducing numerical values gives

$$\phi_{\max} = \frac{\pi \times 10^8}{4 \times 3.2 \times 10^{-11} \times 0.0047 \times (50)^3}$$
$$= 4.18 \times 10^{15}\ \text{neutrons/cm}^2\text{-sec. [}Ans.\text{]}$$

The average value of the flux follows from Eq. (6.42):

$$\phi_{\text{av}} = \frac{\phi_{\max}}{\Omega} = \frac{4.18 \times 10^{15}}{3.29} = 1.27 \times 10^{15}\ \text{neutrons/cm}^2\text{-sec. [}Ans.\text{]}$$

6.4 THE ONE-GROUP CRITICAL EQUATION

It was seen above that a necessary condition for each of the reactors to be critical is that B^2, which was defined in Eq. (6.10) as

$$B^2 = \frac{k_\infty - 1}{L^2},$$

and which is a function of the material properties of the system, must be equal to the square of the first eigenvalue B_1^2, which depends only on the dimensions and geometry of the reactor; in symbols

$$\frac{k_\infty - 1}{L^2} = B_1^2. \tag{6.43}$$

This equation determines the conditions under which a given bare reactor will be critical. If, for instance, the physical properties of the reactor are specified, then the left-hand side of Eq. (6.43) is known and the dimensions of the reactor must be adjusted so that B_1^2 on the right satisfies the equation. On the other hand, if the dimensions are specified, then B_1^2 is known and the properties of the reactor must be adjusted so that the left-hand side of the equation is equal to the right. An example of this procedure follows.

Example 6.3 A fast reactor assembly consisting of a homogeneous mixture of ^{239}Pu and sodium is to be made in the form of a bare sphere. The atom densities of these constituents are $N_F = 0.00395 \times 10^{24}$ for the ^{239}Pu and $N_S = 0.0234 \times 10^{24}$ for the sodium. Estimate the critical radius of the assembly.

Solution. Introducing $B_1^2 = (\pi/R)^2$ from Eq. (6.28) into Eq. (6.43) and solving for R gives

$$R = \pi \sqrt{\frac{L^2}{k_\infty - 1}}.$$

It is necessary, therefore, to compute k_∞ and L^2 in order to find R.

Using the cross sections in Table 6.1 gives

$$\begin{aligned}
\Sigma_{aF} &= 0.00395 \times 2.11 = 0.00833 \text{ cm}^{-1}, \\
\Sigma_{aS} &= 0.0234 \times 0.0008 = 0.000019 \text{ cm}^{-1}, \\
\Sigma_a &= \Sigma_{aF} + \Sigma_{aS} = 0.00835.
\end{aligned}$$

Then from Eq. (6.4)

$$f = \frac{0.00833}{0.00835} \simeq 1,$$

and

$$k_\infty = \eta f \simeq 2.61.$$

To compute $L^2 = D/\Sigma_a$ requires the value of D. This in turn is given by (see Eq. 5.16)

$$D = \frac{1}{3\Sigma_{tr}},$$

where Σ_{tr} is the macroscopic transport cross section. From the values of σ_{tr} given in Table 6.1,

$$\Sigma_{tr} = 0.00395 \times 6.8 + 0.0234 \times 3.3 = 0.104 \text{ cm}^{-1},$$

so that

$$D = \frac{1}{3 \times 0.104} = 3.21 \text{ cm}.$$

Finally,

$$L^2 = \frac{3.21}{0.00835} = 384 \text{ cm}^2.$$

Inserting k_∞ and L^2 into the above expression for R then gives

$$R = \pi \sqrt{\frac{384}{2.61 - 1}} = 48.5 \text{ cm. } [Ans.]$$

The reverse of this problem, namely, the calculation of the critical composition of a reactor whose size is given, is somewhat more complicated, at least for a fast reactor, and will be delayed until the discussion of thermal reactors in the next section.

To return to Eq. (6.43), this may be rearranged and put into the following form,

$$\frac{k_\infty}{1 + B^2L^2} = 1, \tag{6.44}$$

where the subscript denoting the first eigenvalue has been omitted, and where it is understood that, from this point on, B^2 refers to the buckling. Equation (6.44) is known as the one-group *critical equation* for a bare reactor.

It is instructive to examine the physical meaning of the critical equation. For this purpose, consider a critical bare reactor of arbitrary geometry. The number of neutrons absorbed in this reactor per second is $\Sigma_a \int_V \phi \, dV$. On the other hand, the number which leak from the surface of the reactor per second is given by $\int_A \mathbf{J} \cdot \mathbf{n} \, dA$, where $\mathbf{J}$ is the neutron current density at the surface, $\mathbf{n}$ is a unit vector normal to the surface, and the integral is evaluated over the entire surface. From Fick's law and the divergence theorem,

$$\int_A \mathbf{J} \cdot \mathbf{n} \, dA = \int_V \operatorname{div} \mathbf{J} \, dV = -D \int_V \nabla^2\phi \, dV. \tag{6.45}$$

However, from the reactor equation (see Eq. 6.11), this can be written as

$$-D \int_V \nabla^2\phi \, dV = DB^2 \int_V \phi \, dV. \tag{6.46}$$

Neutrons may either leak from the reactor or be absorbed within its interior—there is no other alternative. Therefore, the relative probability, P_L, that a neutron will be absorbed—that is, will *not* leak—is equal to the number absorbed in the reactor per second divided by the sum of that number and the number of neutrons that leak. From the results of the preceding paragraph, it follows that

$$P_L = \frac{\Sigma_a \int_V \phi \, dV}{\Sigma_a \int_V \phi \, dV + DB^2 \int_V \phi \, dV} = \frac{\Sigma_a}{\Sigma_a + DB^2}.$$

Dividing numerator and denominator by Σ_a gives

$$P_L = \frac{1}{1 + B^2L^2}, \tag{6.47}$$

where the definition of L^2 has been used. Equation (6.47) is the one-group formula for the *nonleakage probability* for a bare reactor.

Comparing Eqs. (6.44) and (6.47) shows that the critical equation can be written as

$$k_\infty P_L = 1. \tag{6.48}$$

This result has the following interpretation. A total of $\Sigma_a \int_V \phi\, dV$ neutrons are absorbed in the reactor per second, and this leads to the release of

$$\eta f \Sigma_a \int_V \phi\, dV = k_\infty \Sigma_a \int_V \phi\, dV$$

fission neutrons. Owing to leakage, however, only $P_L k_\infty \Sigma_a \int_V \phi\, dV$ of these are absorbed in the system to initiate a new generation of neutrons. From the definition of k, the multiplication factor of the reactor, it follows that

$$k = \frac{P_L k_\infty \Sigma_a \int_V \phi\, dV}{\Sigma_a \int_V \phi\, dV} = k_\infty P_L. \tag{6.49}$$

Thus the left-hand side of the critical equation is actually the multiplication factor for the reactor, and the critical equation follows by merely placing $k = 1$.

Example 6.4 What is the average probability that a fission neutron in the assembly described in Example 6.3 will be absorbed within the system?

Solution. The probability in question is the nonleakage probability given in Eq. (6.47). Using the values $R = 48.5$ cm and $L^2 = 384$ cm^2 from the example gives

$$P_L = \frac{1}{1 + \left(\dfrac{\pi}{48.5}\right)^2 \times 384} = 0.38. \text{ [}Ans.\text{]}$$

There is thus a 38 percent chance that a neutron will be absorbed, and a 62 percent chance that it will escape.

6.5 THERMAL REACTORS

Thermal reactors are sufficiently distinctive to warrant separate discussion. It will be recalled that these systems contain fuel, coolant, various structural materials, and a moderator to slow down the fission neutrons to thermal energies. For convenience, in this section all the materials in the reactor other than fuel will be called "moderator." In short, the reactor will be assumed to consist only of fuel and moderator.

The Four-Factor Formula

To begin with, consider an infinite reactor composed of a homogeneous fuel-moderator mixture. If $\overline{\Sigma}_a$ is the macroscopic thermal absorption cross section of the mixture, that is,

$$\overline{\Sigma}_a = \overline{\Sigma}_{aF} + \overline{\Sigma}_{aM}, \tag{6.50}$$

where $\overline{\Sigma}_{aF}$ and $\overline{\Sigma}_{aM}$ are the cross sections of the fuel and moderator, respectively, then there will be a total of $\overline{\Sigma}_a \phi_T$ neutrons absorbed per cm^3/sec everywhere in the reactor, where ϕ_T is the thermal flux (see Section 5.9). Of this number, the fraction

$$f = \frac{\overline{\Sigma}_{aF}}{\overline{\Sigma}_a} = \frac{\overline{\Sigma}_{aF}}{\overline{\Sigma}_{aF} + \overline{\Sigma}_{aM}} \tag{6.51}$$

are absorbed by the fuel. The parameter f, which was called the fuel utilization in Section 6.1, is known as the *thermal utilization* in thermal reactors.

There are thus $f\overline{\Sigma}_a \phi_T$ neutrons absorbed per cm^3/sec in fuel, and, as a result, $\eta_T f \overline{\Sigma}_a \phi_T$ fission neutrons are emitted per cm^3/sec, where η_T is the average number of neutrons emitted per thermal neutron absorbed in fuel. The parameter η_T is computed from the integral

$$\eta_T = \frac{\int \eta(E)\sigma_{aF}(E)\phi(E)\, dE}{\int \sigma_{aF}(E)\phi(E)\, dE}, \tag{6.52}$$

where $\phi(E)$ is the Maxwellian flux given by Eq. (5.54). Values of η_T are given in Table 6.3, where it will be observed that η_T is a slowly varying function of temperature.

In thermal reactors containing large amounts of fissionable but nonfissile material such as ^{238}U, a small fraction of the fissions are induced by fast neutrons

TABLE 6.3
Values of η_T, the average number of fission neutrons emitted per neutron absorbed in a thermal flux, at the temperature T

T, °C	^{233}U	^{235}U	^{239}Pu
20	2.284	2.065	2.035
100	2.288	2.063	1.998
200	2.291	2.060	1.947
400	2.292	2.050	1.860
600	2.292	2.042	1.811
800	2.292	2.037	1.785
1000	2.292	2.033	1.770

interacting with the nonfissile nuclides. These fast fissions can be taken into account by introducing a factor ϵ, called the *fast fission factor*, which is defined as the ratio of the total number of fission neutrons produced by both fast and thermal fission to the number produced by thermal fission alone. It follows from this definition that the total number of fission neutrons produced per cm^3/sec in an infinite thermal reactor is equal to $\epsilon\eta_T f\overline{\Sigma}_a\phi_T$. The value of ϵ for reactors fueled with natural or slightly enriched uranium ranges from about 1.02 to 1.08.

As already noted, there can be no leakage of neutrons from an infinite reactor; all of the fission neutrons must eventually be absorbed somewhere in the reactor. In a thermal reactor, most of the neutrons are absorbed after they have slowed down to thermal energies. However some neutrons may be absorbed while slowing down by nuclei having absorption resonances at energies above the thermal region. If p is the probability that a fission neutron is *not* absorbed in any of these resonances, then, of the $\epsilon\eta_T f\overline{\Sigma}_a\phi_T$ fission neutrons produced per cm^3/sec, only $p\epsilon\eta_T f\overline{\Sigma}_a\phi_T$ actually succeed in slowing down to thermal. The parameter p is known as the *resonance escape probability* and is one of the most important factors in the design of a thermal reactor.

From the foregoing discussion it follows that the absorption of $\overline{\Sigma}_a\phi_T$ thermal neutrons leads to the production of $p\epsilon\eta_T f\overline{\Sigma}_a\phi_T$ new thermal neutrons, all of which must eventually be absorbed in an infinite reactor. The absorption of this generation of thermal neutrons leads to another generation of thermal neutrons, and so on. The multiplication factor of the reactor is therefore

$$k_\infty = \frac{p\epsilon\eta_T f\overline{\Sigma}_a\phi_T}{\overline{\Sigma}_a\phi_T} = \eta_T f p\epsilon, \tag{6.53}$$

where the subscript on k_∞ again signifies that the equation is valid only for infinite systems. Since k_∞ is the product of four factors, η_T, f, p, and ϵ, Eq. (6.53) is called the *four-factor formula*. It may be noted that the order of the factors in the right-hand side of Eq. (6.53) has been rearranged to conform with normal usage.

Criticality Calculations

The one-group method, at least in the form given earlier in this chapter, gives only rough estimates of the critical size or composition of a thermal reactor. This is due to the fact that although most of the fission neutrons are ultimately absorbed at thermal energies, they may diffuse about over considerable distances while slowing down. Such fast-neutron diffusion must be taken into consideration. To this end, as discussed in Section 5.10, it is usual to describe a thermal reactor by at least two neutron groups, one group for the fast neutrons—those with energies above the thermal region—and a second group for the thermal neutrons.

In such a two-group calculation, it can usually be assumed that there is no absorption of neutrons in the fast group; resonance absorption is taken into account by the introduction of the resonance escape probability. Neutrons are then lost from the fast group only as the result of scattering into the thermal group. In particular, following the discussion in Section 5.10, there are $\Sigma_1\phi_1$ neutrons scattered per cm^3/sec out of the fast group, where ϕ_1 is the fast flux. Also, in a thermal reactor, it can be assumed that the bulk of the fissions are induced by thermal neutrons; the few fissions induced by fast neutrons are accounted for in the fast fission factor. It follows that $\eta_T f\epsilon\overline{\Sigma}_a\phi_T = (k_\infty/p)\overline{\Sigma}_a\phi_T$ fission neutrons are emitted per cm^3/sec, and these neutrons necessarily appear as source neutrons in the fast group. The source density for the fast group is therefore

$$s_1 = \frac{k_\infty}{p}\overline{\Sigma}_a\phi_T.$$

Substituting this expression into the group diffusion equation (see Eq. 5.50) gives the following equation for the fast group:

$$D_1\nabla^2\phi_1 - \Sigma_1\phi_1 + \frac{k_\infty}{p}\overline{\Sigma}_a\phi_T = 0. \tag{6.54}$$

In the absence of resonance absorption all of the $\Sigma_1\phi_1$ neutrons per cm^3/sec scattered out of the fast group would appear as source neutrons in the thermal flux equation. With resonance absorption present, however, only $p\Sigma_1\phi_1$ neutrons per cm^3/sec actually succeed in entering the thermal group. Thus the thermal source term is

$$s_T = p\Sigma_1\phi_1.$$

Introducing this expression into the thermal diffusion equation then gives

$$\overline{D}\nabla^2\phi_T - \overline{\Sigma}_a\phi_T + p\Sigma_1\phi_1 = 0. \tag{6.55}$$

Equations (6.54) and (6.55) are the two-group equations describing a bare thermal reactor.

It is not difficult to show that in a bare reactor all group fluxes have the same spatial dependence, which is determined by the one-group reactor equation. Thus for a bare thermal reactor the two-group fluxes may be written as

$$\phi_1 = A_1\phi, \tag{6.56}$$

$$\phi_T = A_2\phi, \tag{6.57}$$

where A_1 and A_2 are constants, and where ϕ satisfies the equation

$$\nabla^2\phi + B^2\phi = 0. \tag{6.58}$$

Substituting the last three equations into Eqs. (6.54) and (6.55) yields

$$-(D_1B^2 + \Sigma_1)A_1 + \frac{k_\infty}{p}\overline{\Sigma}_aA_2 = 0 \tag{6.59}$$

and

$$p\Sigma_1A_1 - (\overline{D}B^2 + \overline{\Sigma}_a)A_2 = 0. \tag{6.60}$$

Equations (6.59) and (6.60) are a set of homogeneous linear algebraic equations in the two unknowns A_1 and A_2.

According to Cramer's rule,* Eqs. (6.59) and (6.60) have nontrivial solutions only if the determinant of the coefficients multiplying A_1 and A_2 vanishes, that is, if

$$\begin{vmatrix} -(D_1B^2 + \Sigma_1) & \dfrac{k_\infty}{p}\overline{\Sigma}_a \\ p\Sigma_1 & -(\overline{D}B^2 + \overline{\Sigma}_a) \end{vmatrix} = 0.$$

Multiplying out the determinant gives

$$k_\infty\overline{\Sigma}_a\Sigma_1 - (\overline{D}B^2 + \overline{\Sigma}_a)(D_1B^2 + \Sigma_1) = 0,$$

or, on rearranging,

$$\frac{k_\infty\Sigma_1\overline{\Sigma}_a}{(D_1B^2 + \Sigma_1)(\overline{D}B^2 + \overline{\Sigma}_a)} = 1.$$

Dividing numerator and denominator by $\Sigma_1\overline{\Sigma}_a$ yields finally

$$\frac{k_\infty}{(1 + B^2L_T^2)(1 + B^2\tau_T)} = 1. \qquad (6.61)$$

In this expression

$$L_T^2 = \frac{\overline{D}}{\overline{\Sigma}_a} \qquad (6.62)$$

is the thermal diffusion area discussed in Section 5.9, and τ_T is the parameter called neutron age which was defined in Section 5.10 as

$$\tau_T = \frac{D_1}{\Sigma_1}. \qquad (6.63)$$

*Consider the set of homogeneous linear equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1N}x_N &= 0, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2N}x_N &= 0, \\ \cdots \quad \cdots \quad \cdots \quad & \cdots \\ a_{N1}x_1 + a_{N2}x_2 + \cdots + a_{NN}x_N &= 0, \end{aligned}$$

where a_{mn} are constants. Cramer's rule states that there is no solution other than $x_1 = x_2 = \cdots = x_N = 0$ unless the determinant of the coefficients a_{mn} vanishes, that is,

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1N} \\ a_{21} & a_{22} & \cdots & a_{2N} \\ \cdots & \cdots & \cdots & \cdots \\ a_{N1} & a_{N2} & \cdots & a_{NN} \end{vmatrix} = 0.$$

When this equation is satisfied, the equations in the original set are not independent. That is, although there appear to be N distinct equations in the set, at least one of these equations can be written as a combination of other equations in the set. Since there are N unknowns but only $N - 1$ independent equations, it is possible to determine only $N - 1$ of the x's in terms of the remaining x.

Equation (6.61) is the two-group critical equation for a bare thermal reactor. The factor

$$P_T = \frac{1}{1 + B^2L_T^2} \tag{6.64}$$

in this equation was shown in Section 6.4 to be the probability that a thermal neutron will not leak from the reactor. Also, it is not difficult to show that

$$P_F = \frac{1}{1 + B^2\tau_T} \tag{6.65}$$

is the probability that a fission neutron will not escape from the reactor while slowing down. Therefore, by the argument given in Section 6.4, it follows that the left-hand side of Eq. (6.61) is the multiplication factor of the reactor, that is,

$$k = k_\infty P_T P_F, \tag{6.66}$$

and the critical equation follows by writing $k = 1$.

Since all reactors are designed in such a way that there is as little leakage of neutrons as possible, both P_T and P_F are ordinarily very close to unity. The quantities $B^2L_T^2$ and $B^2\tau_T$ are therefore small, and when the denominator of the critical equation, Eq. (6.61), is multiplied out, the term $B^4L_T^2\tau_T$ can be ignored. The resulting expression is

$$\frac{k_\infty}{1 + B^2(L_T^2 + \tau_T)} = 1,$$

or

$$\frac{k_\infty}{1 + B^2M_T^2} = 1, \tag{6.67}$$

where

$$M_T^2 = L_T^2 + \tau_T \tag{6.68}$$

is called the *thermal migration area.*

It will be observed that Eq. (6.67) is identical in form to the critical equation (Eq. 6.44) in a one-group calculation. For this reason, Eq. (6.67) is known as the *modified one-group* critical equation—modified in the sense that L^2 in the one-group equation has been replaced by M_T^2. Furthermore, from Eqs. (6.57) and (6.58), it follows that the thermal flux is given by the same equation as in a one-group calculation, namely,

$$\nabla^2\phi_T + B^2\phi_T = 0, \tag{6.69}$$

where B^2 is the buckling given in Table 6.2. Thus the only difference between an ordinary one-group calculation and a modified one-group calculation of a bare thermal reactor is that in the latter, L_T^2 is replaced by M_T^2; the flux and the buckling are the same as before.

It may be noted that if τ_T is much less than L_T^2, Eq. (6.67) reduces to the one-group critical equation, Eq. (6.44). In this case the reactor is well described by one-group theory. A comparison of the values of τ_T and L_T^2 in Tables 5.2 and 5.3 shows that, with the exception of water, τ_T is in fact less than L_T^2 for the common moderators; it is much less than L_T^2 for D_2O and graphite. It would be quite wrong, therefore, to use an ordinary one-group calculation, omitting τ_T, for a water moderated reactor; somewhat less error would be involved in such a computation of a D_2O or graphite moderated system, provided the value of L_T^2, with fuel mixed in with the moderator, is still substantially larger than τ_T.

Applications

The above results will now be applied to the practical problem of determining the critical composition or critical dimensions of a bare thermal reactor. For simplicity, the discussion will be limited to reactors that do not contain resonance absorbers or nuclei that undergo fast fission. The reactor, in short, is assumed to consist of a homogeneous mixture of a fissile isotope and moderator. In this case $p = \epsilon = 1$. Then according to Eq. (6.53) k_∞ reduces to

$$k_\infty = \eta_T f. \tag{6.70}$$

Reactors with resonance absorption and fast fission are treated in Section 6.8.

It will be recalled that there are two situations to be considered: (1) the physical size of the reactor is specified and its critical composition must be determined, or (2) the composition of the reactor is specified and its critical size must be determined. These problems will now be taken up in turn.

Case 1. Size Specified. With size given, B^2 can be immediately computed from the appropriate formula in Table 6.2. The composition must then be adjusted so that k_∞ and M_T^2 have the values necessary to satisfy the critical equation, Eq. (6.67).

It is first convenient to introduce the parameter Z defined by

$$Z = \frac{\overline{\Sigma}_{aF}}{\overline{\Sigma}_{aM}} = \frac{N_F\bar{\sigma}_{aF}}{N_M\bar{\sigma}_{aM}}, \tag{6.71}$$

where, as usual, the subscripts F and M denote the fuel and moderator, respectively. Then from Eq. (6.51) the thermal utilization can be written as

$$f = \frac{Z}{Z+1}, \tag{6.72}$$

and, in view of Eq. (6.70), k_∞ takes the form

$$k_\infty = \frac{\eta_T Z}{Z+1}. \tag{6.73}$$

Consider next the thermal diffusion area, which is one of the two factors entering into M_T^2 (see Eq. 6.68). From Eq. (6.62)

$$L_T^2 = \frac{\overline{D}}{\overline{\Sigma}_a},$$

where $\overline{D}$ and $\overline{\Sigma}_a$ refer to the homogeneous mixture of fuel and moderator. However, $\overline{D}$ is essentially equal to $\overline{D}_{\mathrm{M}}$, the diffusion coefficient for the moderator, since the concentration of the fuel in the moderator is ordinarily small for homogeneous thermal reactors. Thus L_T^2 becomes

$$L_T^2 = \frac{\overline{D}_{\mathrm{M}}}{\overline{\Sigma}_a} = \frac{\overline{D}_{\mathrm{M}}}{\overline{\Sigma}_{a\mathrm{F}} + \overline{\Sigma}_{a\mathrm{M}}}.$$

Dividing numerator and denominator by $\overline{\Sigma}_{a\mathrm{M}}$ and using the definition of Z from Eq. (6.71) gives

$$L_T^2 = \frac{L_{T\mathrm{M}}^2}{Z+1}, \tag{6.74}$$

where $L_{T\mathrm{M}}^2$ is the thermal diffusion area of the moderator. Upon solving Eq. (6.72) for Z and substituting into Eq. (6.74), the result is

$$L_T^2 = (1 - f)L_{T\mathrm{M}}^2. \tag{6.75}$$

The age τ_T, like $\overline{D}$, depends primarily on the scattering properties of the medium. In the homogeneous type of reactor under consideration, very little fissile material is necessary to reach criticality. Furthermore, the scattering cross sections of fissile atoms are not significantly larger than those of the ordinary moderators. It is possible, therefore, to ignore the presence of the fuel altogether, and use for τ_T the value of the age for the moderator alone, $\tau_{T\mathrm{M}}$, which is given in Table 5.3.

Introducing Eqs. (6.73) and (6.74) into the critical equation (Eq. 6.67) then gives

$$\frac{\eta_T Z}{Z + 1 + B^2(L_{T\mathrm{M}}^2 + Z\tau_{T\mathrm{M}} + \tau_{T\mathrm{M}})} = 1.$$

When this equation is solved for Z, the result is found to be

$$Z = \frac{1 + B^2(L_{T\mathrm{M}}^2 + \tau_{T\mathrm{M}})}{\eta_T - 1 - B^2\tau_{T\mathrm{M}}}. \tag{6.76}$$

This is the value of Z which leads to a critical reactor with the specified value of B^2.

To find, using this value of Z, the total mass of fuel required for criticality, that is, the *critical mass*, the procedure is as follows. From Eq. (6.71) the atom density of the fuel is

$$N_{\mathrm{F}} = Z\frac{\overline{\sigma}_{a\mathrm{M}}}{\overline{\sigma}_{a\mathrm{F}}}N_{\mathrm{M}}. \tag{6.77}$$

The *total* number of fuel atoms in the reactor is $N_F V$, where V is the reactor volume, and the total number of moles of fuel is $N_F V/N_A$, where N_A is Avogadro's number. Then if M_F is the gram atomic weight of the fuel, it follows that m_F, the fuel mass, is given by

$$m_F = \frac{N_F V M_F}{N_A} \tag{6.78}$$

$$= Z\frac{\bar{\sigma}_{aM} V M_F}{\bar{\sigma}_{aF} N_A} N_M. \tag{6.79}$$

However, the total mass of the moderator is

$$m_M = \frac{N_M V M_M}{N_A},$$

so that Eq. (6.79) can also be written as

$$m_F = Z\frac{\bar{\sigma}_{aM} M_F}{\bar{\sigma}_{aF} M_M} m_M. \tag{6.80}$$

Introducing the formulas given in Chapter 5 for $\bar{\sigma}_{aM}$ and $\bar{\sigma}_{aF}$ (see Eq. 5.64) gives finally

$$m_F = Z\frac{\sigma_{aM}(E_0) M_F}{g_{aF}(T)\sigma_{aF}(E_0) M_M} m_M, \tag{6.81}$$

where the non-$1/v$ factor of the moderator has been taken to be unity and the cross sections are evaluated at the energy $E_0 = 0.0253$ eV. Because the fuel concentration is normally so small, the moderator mass can be computed using its ordinary density as shown in the following example.

Example 6.5 A bare spherical thermal reactor, 100 cm in radius, consists of a homogeneous mixture of ^{235}U and graphite. The reactor is critical and operates at a power level of 100 thermal kilowatts. Using modified one group theory, calculate (a) the buckling; (b) the critical mass; (c) k_∞; (d) L_T^2; (e) the thermal flux. For simplicity, make all computations at room temperature.

Solution.

a) From Table 6.2, $B^2 = (\pi/R)^2 = (\pi/100)^2 = 9.88 \times 10^{-4}$ cm^{-2}, and $B = 3.14 \times 10^{-2}$. [*Ans.*]

b) According to Tables 5.2, 5.3, and 6.3, $L_{TM}^2 = 3500$ cm^2, $\eta_T = 2.065$, and $\tau_{TM} = 368$ cm^2. Then from Eq. (6.76),

$$Z = \frac{1 + 9.88 \times 10^{-4}(3500 + 368)}{2.065 - 1 - 9.88 \times 10^{-4} \times 368} = 6.87.$$

Using the values $\sigma_{aM}(E_0) = 0.0034$ b, $\sigma_{aF}(E_0) = 681$ b, and $g_{aF}(T) = 0.978$ in Eq. (6.81) gives

$$m_F = \frac{6.87 \times 0.0034 \times 235}{0.978 \times 681 \times 12} m_M = 6.87 \times 10^{-4} m_M.$$

The density of graphite is approximately 1.60 g/cm^3, so that the total mass of graphite in the reactor is $m_M = \frac{4}{3}\pi R^3 \times 1.60 = 6.70 \times 10^6$ g $= 6700$ kg. It follows that the critical mass is $m_F = 6.87 \times 10^{-4} \times 6700 = 4.60$ kg. [*Ans.*]

c) From Eq. (6.72),

$$f = \frac{Z}{Z+1} = \frac{6.87}{7.87} = 0.873,$$

and

$$k_\infty = \eta_T f = 2.065 \times 0.873 = 1.803. \text{ [}Ans.\text{]}$$

d) From Eq. (6.75),

$$L_T^2 = (1 - f)L_{TM}^2 = (1 - 0.873) \times 3500 = 444 \text{ cm}^2. \text{ [}Ans.\text{]}$$

e) The thermal flux is given by

$$\phi_T = A\frac{\sin Br}{r},$$

where, according to Table 6.2, $A = P/4R^2E_R\overline{\Sigma}_f$. Here $P = 100$ kW $= 10^5$ joules/sec and $E_R = 3.2 \times 10^{-11}$ joule. The value of $\overline{\Sigma}_f = N_F\overline{\sigma}_f$ can be found by noting from Eq. (6.78) that

$$N_F = \frac{m_F N_A}{VM_F},$$

so that

$$\overline{\Sigma}_f = \frac{m_F N_A \overline{\sigma}_f}{VM_F}$$
$$= \frac{m_F N_A}{VM_F} \times 0.886 g_{fF}(T)\sigma_f(E_0).$$

Using the values $g_{fF}(T) = 0.976$ and $\sigma_f(E_0) = 582$ b gives $\overline{\Sigma}_f = 1.41 \times 10^{-3}$ cm^{-1}. The constant A is then

$$A = \frac{10^5}{4 \times 10^4 \times 3.2 \times 10^{-11} \times 1.41 \times 10^{-3}} = 5.54 \times 10^{13}$$

and the flux is

$$\phi_T(r) = 5.54 \times 10^{13}\frac{\sin Br}{r}. \text{ [}Ans.\text{]}$$

The maximum value of ϕ_T occurs at $r = 0$ and is equal to $\phi_T(0) = 5.54 \times 10^{13}B = 1.74 \times 10^{12}$ neutrons/cm^2-sec.

Case 2. Composition Specified. When the composition is given and the critical dimensions must be found, the parameters k_∞ and M_T^2 can be computed directly. The value of B^2 can then be obtained from Eq. (6.67), namely,

$$B^2 = \frac{k_\infty - 1}{M_T^2}. \tag{6.82}$$

If the geometry of the reactor is specified, the dimensions can then be determined from the appropriate formula for B^2 in Table 6.2. Thus for a cubical reactor of side a, $a = \pi\sqrt{3}/B$, while for a sphere the critical radius is $R = \pi/B$.

If, however, it is only specified that the reactor is, say, a finite cylinder, then the buckling formula in the table only provides a relationship between its height H and radius R that must be satisfied if the reactor is critical. Each combination of H and R leads to a different reactor volume and hence a different critical mass. It is not difficult to show that this mass is smallest when H and R satisfy the relationship $H = 1.82R$. The cross section through the axis of such a cylinder is almost square. The parallelepiped with least volume, as might be expected, is a cube.

Example 6.6 A 5-watt experimental thermal reactor is constructed in the form of a cylinder. The reactor is fueled with a homogeneous mixture of ^{235}U and ordinary water with a fuel concentration of 0.0145 g/cm^3. Because of its low power, the system operates at essentially room temperature and atmospheric pressure. (a) Calculate the dimensions of the cylinder which has the smallest critical mass. (b) Determine the critical mass.

Solution.

a) The ratio of the number of atoms of ^{235}U per cm^3 to water molecules per cm^3 is

$$\frac{N_F}{N_M} = \frac{\rho_F M_M}{\rho_M M_F},$$

where ρ_F and ρ_M are the densities of ^{235}U and water in the mixture, respectively, and M_F and M_M are their gram atomic and molecular weights. Introducing these values gives

$$\frac{N_F}{N_M} = \frac{0.0145 \times 18}{1 \times 235} = 1.11 \times 10^{-3}.$$

The average thermal absorption cross sections of fuel and moderator are

$$\bar{\sigma}_{aF} = 0.886 g_{aF}(20°C)\sigma_{aF}(E_0) = 0.886 \times 0.978 \times 681 = 590 \text{ b}$$

and

$$\bar{\sigma}_{aM} = 0.886 \times 2 \times \sigma_{aH}(E_0) = 0.886 \times 2 \times 0.332 = 0.588 \text{ b}.$$

Then from Eq. (6.71),

$$Z = \frac{N_F \bar{\sigma}_{aF}}{N_M \bar{\sigma}_{aM}} = 1.11 \times 10^{-3} \times \frac{590}{0.588} = 1.11.$$

The thermal utilization is

$$f = \frac{Z}{Z+1} = \frac{1.11}{2.11} = 0.526$$

and

$$k_\infty = \eta_T f = 2.065 \times 0.526 = 1.0862.$$

According to Table 5.2, L_{TM}^2 for water is 8.1 cm^2, so that from Eq. (6.75), L_T^2 for the fuel-moderator mixture is

$$L_T^2 = (1 - f)L_{TM}^2 = (1 - 0.526) \times 8.1 = 3.84 \text{ cm}^2.$$

The neutron age from Table 5.3 is 27 cm^2, and hence

$$M_T^2 = L_T^2 + \tau_T = 3.84 + 27 = 30.8 \text{ cm}^2.$$

Then from Eq. (6.82)

$$B^2 = \frac{k_\infty - 1}{M_T^2} = \frac{1.0862 - 1}{30.8} = 2.80 \times 10^{-3} \text{ cm}^{-2}.$$

From Table 6.2, $B^2 = (2.405/R)^2 + (\pi/H)^2$. However, for minimum critical mass it is necessary that $H = 1.82R$. This means that

$$B^2 = \left(\frac{2.405}{R}\right)^2 + \left(\frac{\pi}{1.82R}\right)^2 = \frac{8.763}{R^2}.$$

Solving for R^2 gives

$$R^2 = \frac{8.763}{B^2} = \frac{8.763}{2.80 \times 10^{-3}} = 3.13 \times 10^3.$$

Thus, $R = 55.9$ cm and $H = 101.7$ cm. [*Ans.*]

b) The reactor volume is $\pi R^2 H = \pi \times (55.9)^2 \times 101.7 = 9.98 \times 10^5$ cm^3. Since there are 0.0145 g/cm^3 of ^{235}U, the fuel mass is

$$m_F = 0.0145 \times 9.98 \times 10^5 = 1.45 \times 10^4 \text{ g} = 14.5 \text{ kg}. \text{ [}Ans.\text{]}$$

6.6 REFLECTED REACTORS

It was pointed out in Chapter 4 that the neutron economy is improved when the core of a reactor is surrounded by a *reflector*, that is, by a thick, unfueled region of moderator. The neutrons that otherwise would leak from the bare core now pass into the reflector, and some of these diffuse back into the core. The net result is that the critical mass of the system is reduced.

Criticality calculations for reflected reactors will now be considered within the framework of one-group diffusion theory. It will be recalled that this method is applicable to calculations of fast reactors and thermal reactors, such as those moderated by D_2O or graphite, for which $\tau_T \ll L_T^2$. Reflected water reactors for which $\tau_T \gg L_T^2$ will be treated separately later in this section.

As a specific example, consider a spherical reactor consisting of a core of radius R, surrounded by an infinite reflector. In the analysis which follows, parameters which refer to the core and reflector will be denoted by subscripts c and r, respectively.

According to one-group theory, the flux in the core ϕ_c satisfies the equation (see Eq. 6.11)

$$\nabla^2\phi_c + B^2\phi_c = 0, \tag{6.83}$$

where

$$B^2 = \frac{k_\infty - 1}{L_c^2}. \tag{6.84}$$

Since there is no fuel in the reflector, the flux in this region satisfies the one-group diffusion equation

$$\nabla^2\phi_r - \frac{1}{L_r^2}\phi_r = 0. \tag{6.85}$$

In order to find the flux throughout the reactor and the conditions for criticality, it is necessary to solve Eqs. (6.83) and (6.85) for ϕ_c and ϕ_r subject to boundary conditions.

The general solution to Eq. (6.83) was derived in Section 6.3, and is

$$\phi_c = A\frac{\sin Br}{r} + C\frac{\cos Br}{r},$$

where A and C are constants. Since ϕ_c must not be infinite at the center of the reactor (at $r = 0$), it is necessary to place $C = 0$, so that ϕ_c reduces to

$$\phi_c = A\frac{\sin Br}{r} \tag{6.86}$$

The general solution to Eq. (6.85) is

$$\phi_r = A'\frac{e^{-r/L_r}}{r} + C'\frac{e^{r/L_r}}{r},$$

where A' and C' are constants. However, ϕ_r must remain finite as r goes to infinity, and C' must therefore be taken to be zero. The flux in the reflector is then

$$\phi_r = A' \frac{e^{-r/L_r}}{r}. \tag{6.87}$$

The functions ϕ_c and ϕ_r must also satisfy the interface boundary conditions (see Section 5.5), namely, continuity of neutron flux and current at the core-reflector interface, that is, at $r = R$. Written out in detail, these conditions are

$$\phi_c(R) = \phi_r(R) \tag{6.88}$$

and

$$J_c(R) = J_r(R)$$

or

$$D_c\phi_c'(R) = D_r\,\phi_r'(R), \tag{6.89}$$

where the prime indicates differentiation with respect to r.

Introducing Eqs. (6.86) and (6.87) into Eq. (6.88) gives

$$A\,\frac{\sin BR}{R} = A'\,\frac{e^{-R/L_r}}{R}. \tag{6.90}$$

Next, differentiating Eqs. (6.86) and (6.87) and inserting the results into Eq. (6.89) yields

$$AD_c\left(\frac{B\cos BR}{R} - \frac{\sin BR}{R^2}\right) = -A'D_r\left(\frac{1}{RL_r} + \frac{1}{R^2}\right)e^{-R/L_r}. \tag{6.91}$$

Equations (6.90) and (6.91) are homogeneous linear equations in the unknowns A and A'; they have nontrivial solutions only if the determinant of the coefficients vanishes. Multiplying out this determinant (which is equivalent to dividing one equation by the other) gives

$$D_c\left(B\cot BR - \frac{1}{R}\right) = -D_r\left(\frac{1}{L_r} + \frac{1}{R}\right).$$

For computational purposes it is convenient to rearrange this equation in the following form:

$$BR\cot BR - 1 = -\frac{D_r}{D_c}\left(\frac{R}{L_r} + 1\right). \tag{6.92}$$

Equation (6.92) must be satisfied for the reactor to be critical. If, for example, the composition of the core is given, then B is known from Eq. (6.84), and R can be calculated from Eq. (6.92). On the other hand, if R is specified, B must be computed from Eq. (6.92), and the critical composition of the core can be determined by using Eq. (6.84). The only complicating feature of these calculations which was not present with bare reactors is that Eq. (6.92) is a transcendental equation.

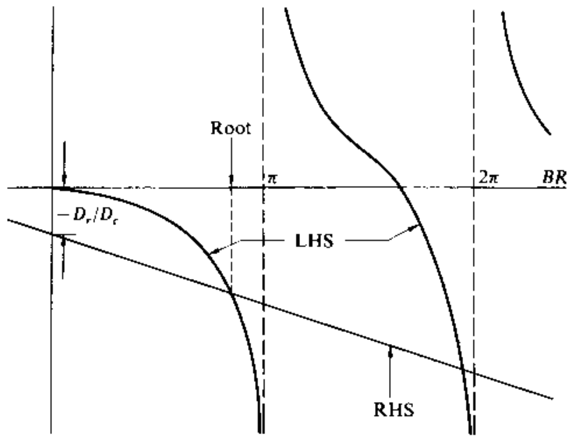

Fig. 6.4 A plot of the two sides of Eq. (6.92).

It is instructive to consider the solution of Eq. (6.92) graphically. Suppose, for instance, that the composition of the core, and hence B, is specified. The critical core radius can then be found by plotting the left-hand side (LHS) and the right-hand side (RHS) of Eq. (6.92) separately as functions of BR. This is shown in Fig. 6.4. The LHS has an infinite number of branches each of width π, while the RHS is a straight line of slope $-D_r/D_cBL_r$* which intersects the vertical axis at $-D_r/D_c$. Every value of BR corresponding to an intersection of the LHS and RHS is a root or solution to Eq. (6.92), and since there are an infinite number of intersections, there are also an infinite number of solutions. However, as explained earlier in this chapter, only the first solution is relevant to a critical reactor, the higher roots corresponding to solutions which die out in time. With the value of BR obtained from the first intersection and with B known, the critical radius can be immediately determined.

It should be noted in Fig. 6.4 that the first intersection of LHS and RHS occurs at a value of BR which is less than π. Had the reactor been bare instead of reflected, BR would have been exactly equal to π. It follows, therefore, that the critical core radius is less for the reflected reactor than for the bare reactor of the same composition—a conclusion that had been anticipated on physical grounds.

Incidentally, it may be noted that, in the special case in which the moderator in the core and reflector are the same, $D_r = D_c$, and Eq. (6.92) reduces to

$$B \cot BR = -\frac{1}{L_r}. \tag{6.93}$$

*As a function of BR,

$$\text{RHS} = -\frac{D_r}{D_c}\left(\frac{BR}{BL_r} + 1\right).$$

This equation is not transcendental in R. Thus if B is known, R can be calculated directly. This case is illustrated in Example 6.8 below.

Having obtained the condition for criticality, it is now necessary to return to Eqs. (6.90) and (6.91) and find the values of the constants A and A' that determine the flux. These equations are not independent, however, but are related through Eq. (6.92). It is possible, therefore, to find A' in terms of A, or A in terms of A', but both constants cannot be found independently. For example, from Eq. (6.90), A' is

$$A' = Ae^{R/L_r} \sin BR, \tag{6.94}$$

where BR is the known solution to Eq. (6.92).

The constant A, which fixes the magnitude of the flux throughout the reactor, is determined by the reactor power. As in Section 6.3, this is given by

$$P = E_R \Sigma_f \int \phi_c \, dV, \tag{6.95}$$

where the integral extends over the reactor core—the power-producing region. Inserting ϕ_c from Eq. (6.86) and using the volume element $dV = 4\pi r^2\, dr$ gives

$$\begin{aligned} P &= 4\pi E_R \Sigma_f A \int_0^R r \sin Br \, dr \\ &= \frac{4\pi E_R \Sigma_f A}{B^2} (\sin BR - BR \cos BR). \end{aligned}$$

When this is solved for A, the result is

$$A = \frac{PB^2}{4\pi E_R \Sigma_f (\sin BR - BR \cos BR)}. \tag{6.96}$$

The above procedure—solving the core and reflector equations, obtaining an equation that must be satisfied for criticality, and evaluating the flux constants in terms of the reactor power—can be carried through analytically for only a few reactor geometries. Unfortunately, these do not include the two most important reactors, namely, the fully reflected parallelepiped and cylinder. In practice, the properties of these reactors are often estimated from calculations of spherical reactors of the same composition and volume.

It may be noted that the analysis of reflected reactors was carried out in this section without regard for whether the reactors were fast or thermal. The various formulas derived above are therefore valid for both types of systems. To use these formulas in calculations of thermal reactors it is merely necessary to replace each reactor parameter by its appropriate thermal averaged value. The one-group flux, of course, is then the thermal flux. This is illustrated in the next example.

Example 6.7 The core of a thermal reactor is a sphere 300 cm in radius which contains a homogeneous mixture of ^{235}U and D_2O. This is surrounded by an infinite graphite reflector. Calculate for room temperature (a) the critical concentration of ^{235}U in grams per liter; (b) the critical mass.

Solution.

a) To find the fuel concentration, the value of B must first be found from Eq. (6.92). Introducing the values from Table 5.2, $\overline{D}_r = 0.84$ cm, $\overline{D}_c = 0.87$ cm, $R = 300$ cm, and $L_r = L_{Tr} = 59$ cm, gives

$$BR \cot BR = 1 - \frac{0.84}{0.87}\left(\frac{300}{59} + 1\right) = -4.88.$$

Solving, graphically or otherwise, for BR yields $BR = 2.64$ and

$$B = \frac{2.64}{300} = 8.80 \times 10^{-3}\ \text{cm}^{-1}.$$

Next, letting

$$Z = \frac{N_F\,\bar{\sigma}_{aF}}{N_M\bar{\sigma}_{aM}},$$

it follows as in Section 6.5 that

$$k_\infty = \frac{\eta_T Z}{Z + 1}$$

and

$$L_T^2 = (1 - f)L_{TM}^2 = L_{TM}^2/(Z + 1).$$

Inserting these expressions into the definition of B^2, specifically

$$B^2 = \frac{k_\infty - 1}{L_T^2},$$

and solving for Z gives

$$Z = \frac{1 + B^2L_{TM}^2}{\eta_T - 1}.$$

Introducing the values $B = 8.80 \times 10^{-3}$ cm^{-1}, $L_{TM}^2 = 9.4 \times 10^3$ cm^2, and $\eta_T = 2.065$ then gives

$$Z = \frac{1 + (8.80 \times 10^{-3})^2 \times 9.4 \times 10^3}{2.065 - 1} = 1.622$$

From the definition of Z and the usual formula for computing atom density, it is easy to see that the ratio of the physical densities of fuel to moderator is given by

$$\frac{\rho_F}{\rho_M} = Z\,\frac{M_F\bar{\sigma}_{aM}}{M_M\bar{\sigma}_{aF}} = Z\,\frac{M_F\sigma_{aM}(E_0)}{M_M g_{aF}(20°\text{C})\sigma_{aF}(E_0)},$$

where M_M is the molecular weight of D_2O and the absorption cross sections are taken at $E_0 = 0.0253$ eV. Using the values $M_F = 235$, $\sigma_{aM}(E_0) =$

$(\bar{\Sigma}_{aM}/N) \times \sqrt{4/\pi} = 9.3 \times 10^{-5} \times 1.128/.03323 = 3.16 \times 10^{-3}$ (taking into account the 0.25 % H_2O in D_2O), $M_M = 20$, $g_{aF}(20°C) = 0.978$, and $\sigma_{aF}(E_0) = 681$ b, gives

$$\frac{\rho_F}{\rho_M} = 1.622 \times \frac{235 \times 3.16 \times 10^{-3}}{20 \times 0.978 \times 681} = 9.04 \times 10^{-5}.$$

The density of D_2O is 1.1 g/cm^3 so that

$$\rho_F = 9.04 \times 10^{-5} \times 1.1 = 9.94 \times 10^{-5}\ \text{g/cm}^3 = 0.0994\ \text{g/li.}\ [\textit{Ans.}]$$

b) The critical mass is

$$m_F = V\rho_F = \tfrac{4}{3}\pi R^3 \rho_F = \tfrac{4}{3} \times \pi \times (300)^3 \times 9.94 \times 10^{-5}$$
$$= 1.124 \times 10^4\ \text{g} = 11.24\ \text{kg.}\ [\textit{Ans.}]$$

Example 6.8 A large spherical thermal reactor, moderated and reflected by an infinite reflector of graphite, is fueled with ^{235}U at a concentration of 2×10^{-4} g/cm^3. (a) Calculate the critical radius of the core. (b) Compute what the critical radius would be if the reactor were bare. Make calculations at room temperature.

Solution.

a) Since the moderator and reflector are the same, the radius of the reactor is determined by Eq. (6.93),

$$B \cot BR = -\frac{1}{L_{Tr}},$$

where L_{Tr} is the thermal diffusion length of the reflector and B is found from

$$B^2 = \frac{k_\infty - 1}{L_{Tc}^2}.$$

First, it is convenient to compute Z:

$$Z = \frac{N_F\bar{\sigma}_{aF}}{N_M\bar{\sigma}_{aM}} = \frac{\rho_F M_M \bar{\sigma}_{aF}}{\rho_M M_F \bar{\sigma}_{aM}} = \frac{\rho_F M_M g_{aF}(20°C)\sigma_{aF}(E_0)}{\rho_M M_F \sigma_{aM}(E_0)}.$$

Substituting $\rho_F = 2 \times 10^{-4}$ g/cm^3, $\rho_M = 1.6$ g/cm^3, $M_F = 235$, $M_M = 12$, $g_{aF}(20°C) = 0.978$, $\sigma_{aF}(E_0) = 681$ b, $\sigma_{aM} = 3.4 \times 10^{-3}$ b, gives

$$Z = \frac{2 \times 10^{-4} \times 12 \times 0.978 \times 681}{1.6 \times 235 \times 3.4 \times 10^{-3}} = 1.25.$$

Then

$$f = \frac{Z}{Z+1} = 0.556,$$
$$k_\infty = 2.065 \times 0.556 = 1.148,$$
$$L_{Tc}^2 = (1 - f)L_{TM}^2 = (1 - 0.556) \times 3500 = 1554\ \text{cm}^2,$$

and

$$B^2 = \frac{1.148 - 1}{1554} = 9.52 \times 10^{-5},$$

so that $B = 9.76 \times 10^{-3}\ \mathrm{cm}^{-1}$.

From Eq. (6.93),

$$\cot BR = -\frac{1}{BL_{Tr}} = -\frac{1}{9.76 \times 10^{-3} \times 59} = -1.74,$$
$$BR = 2.62,$$

and finally

$$R = \frac{2.62}{B} = \frac{2.62}{9.76 \times 10^{-3}} = 268\ \mathrm{cm}.\ [Ans.]$$

b) Had the reactor been unreflected, the radius R_0 would be given by

$$B = \frac{\pi}{R_0}$$

or

$$R_0 = \frac{\pi}{B} = \frac{\pi}{9.76 \times 10^{-3}} = 322\ \mathrm{cm}.\ [Ans.]$$

Flux in a Reflected Thermal Reactor

Although the one-group method may provide reasonable values for the critical mass, this method does not accurately predict the flux throughout the reactor, especially in the case of a thermal reactor. To do so requires a two-group or multigroup calculation. While such computations for a reflected reactor are beyond the scope of this book, the results are interesting and have an important bearing on the design of a thermal reactor.

To be specific, consider a two-group calculation of a reflected system. As usual, the fast flux, ϕ_1, describes the behavior of the fast neutrons; the thermal neutrons are included in the thermal flux, ϕ_T. One of the striking results of such a computation is that the thermal flux is found to rise near the core-reflector interface and exhibits a peak in the reflector as shown in Fig. 6.5. This behavior of ϕ_T, which is not predicted in one-group theory, is due to the thermalization in the reflector of fast neutrons escaping from the core. These thermalized neutrons are not absorbed as quickly in the reflector as neutrons thermalizing in the core, since the reflector, being unfueled, has a much smaller absorption cross section. The thermal neutrons therefore tend to accumulate in the reflector until they leak back into the core, escape from the outer surface of the reflector, or are absorbed.

An important consequence of the rise in the thermal flux in the reflector is that it tends to flatten the thermal flux distribution in the core. This effect can be seen in Fig. 6.5, which also shows the flux for the bare reactor. The flux in the core

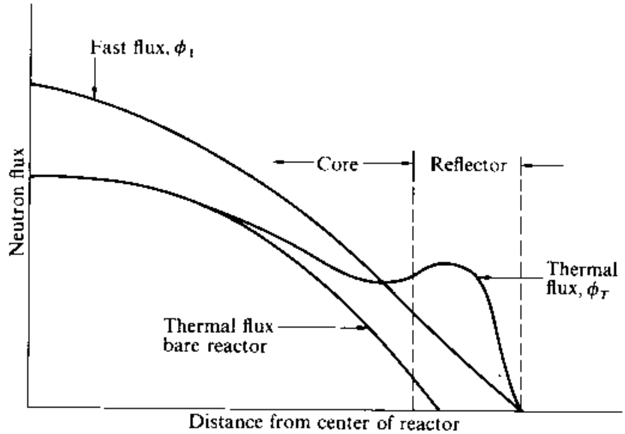

Fig. 6.5 Fast and thermal fluxes in a reflected thermal reactor and the thermal flux in the equivalent reactor.

near the core-reflector interface is built up by the flow of thermal neutrons out of the reflector. Thus a reflector not only reduces the critical size and mass of a reactor, it also reduces the maximum to average flux ratio.

Reflector Savings

The decrease in the critical dimensions of a reactor core as the result of the use of a reflector is called the *reflector savings*. Thus, for a spherical reactor, if R_0 is the radius of the bare core and R is the core radius with the reflector present, the reflector savings is defined as

$$\delta = R_0 - R. \tag{6.97}$$

For instance, for the reactor in Example 6.8, the reflected core radius is 268 cm, while the radius of a bare reactor of the same composition is 322 cm. The reflector savings in this case is $\delta = 322 - 268 = 54$ cm.

For a slab reactor of unreflected and reflected core thicknesses a_0 and a, respectively, the reflector savings is defined as

$$\delta = \tfrac{1}{2}(a_0 - a), \tag{6.98}$$

where the factor ½ is necessary because the slab has two sides.

If the reflector savings is known beforehand, the calculation of the critical dimensions of a reflected reactor requires only the solution of the far simpler problem of the bare reactor. Thus for the reflected sphere, it is only necessary to determine the bare critical radius R_0; the reflected radius is then simply $R = R_0 - \delta$. Of course, a knowledge of δ itself requires either an analytical or experimental solution to the reflected reactor problem. The practical importance of the reflector savings lies in the fact that δ is relatively insensitive to changes in the composition of the reactor. This means that if δ has been determined for one reactor, then the same value of δ can be used for a different reactor of similar composition.

For purposes of making rough calculations of reflected dimensions and/or critical masses of thermal reactors, except those moderated or reflected with water, the following simple formula can be used to estimate δ:

$$\delta \simeq \frac{\overline{D}_c}{\overline{D}_r} L_{\mathrm{Tr}}, \tag{6.99}$$

where the symbols have their usual meanings. Equation (6.99) is only valid provided the reflector is several diffusion lengths thick. The reflector, in this case, is effectively infinite; that is, an increase in reflector thickness does not reduce the critical size of the core. In practice, virtually all reactors have infinite reflectors.

As mentioned earlier, the one-group method, employed in this section to treat reflected reactors, cannot accurately be applied to water moderated and reflected systems owing to the fact that τ_T is much larger than L_T^2. However, to handle these types of reactors, the following empirical formula, developed by R. W. Deutsch, may be used to obtain the reflector savings:

$$\delta = 7.2 + 0.10(M_T^2 - 40.0), \tag{6.100}$$

where δ is in cm, and M_T^2, the migration area of the core, is in cm^2. The use of Eq. (6.100) is illustrated by the following example.

Example 6.9 Calculate the critical core radius and critical mass of a spherical reactor, moderated and reflected by unit-density water. The core contains ^{235}U at a concentration of 0.0145 g/cm^3 (same as in Example 6.6).

Solution. From Example 6.6, $M_T^2 = 30.8$ cm^2 and so

$$\delta = 7.2 + 0.10(30.8 - 40.0) = 6.3 \text{ cm}.$$

Also from that example, $B^2 = 2.80 \times 10^{-3}$ cm^{-2} so that $B = 0.0529$. The bare radius is given by

$$R_0 = \frac{\pi}{B} = \frac{\pi}{0.0529} = 59.4 \text{ cm},$$

and the reflected radius is

$$R = R_0 - \delta = 59.4 - 6.3 = 53.1 \text{ cm. } [Ans.]$$

The critical mass is then

$$\begin{aligned} m_{\mathrm{F}} &= 0.0145 \times \tfrac{4}{3} \times \pi \times (53.1)^3 = 9.09 \times 10^3 \text{ g} \\ &= 9.09 \text{ kg. } [Ans.] \end{aligned}$$

This is considerably less than the 14.5 kg for the unreflected reactor of Example 6.6.

6.7 MULTIGROUP CALCULATIONS

The one-group method gives only the roughest estimates of the properties of a critical reactor, and in order to obtain more accurate results it is necessary to perform multigroup calculations of the type described in Section 5.8. To set up the multigroup equations for a critical reactor, three new group constants must first be defined:

Σ_{fg} = the group-averaged macroscopic fission cross section;
ν_g = the average number of fission neutrons released as the result of fissions induced by neutrons in the gth group; and
χ_g = the fraction of fission neutrons which are emitted with energies in the gth group.

The number of fissions per cm^3/sec in the hth group can now be written as $\Sigma_{fh}\phi_h$, where ϕ_h is the flux in the hth group. As a result of these fissions, $\nu_h\Sigma_{fh}\phi_h$ neutrons are released. The total number of neutrons emitted as the result of fissions in *all* groups is then $\sum \nu_h\Sigma_{fh}\phi_h$, where the sum is carried out over all groups. If the fraction χ_g of these neutrons appears in the gth group, it follows that the source term s_g for the gth multigroup equation (Eq. 5.50) is

$$s_g = \chi_g \sum_{h=1}^{N} \nu_h\Sigma_{fh}\phi_h. \tag{6.101}$$

Introducing this expression into Eq. (5.50) then gives

$$\begin{aligned} D_g\nabla^2\phi_g - \Sigma_{ag}\phi_g - \sum_{h=g+1}^{N} \Sigma_{g\to h}\phi_g + \sum_{h=1}^{g-1} \Sigma_{h\to g}\phi_h& \\ + \chi_g \sum_{h=1}^{N} \nu_h\Sigma_{fh}\phi_h = 0.& \end{aligned} \tag{6.102}$$

This is the gth-group equation in a set of N multigroup equations.

For the usual reactor, which has several regions of differing material properties, there is a set of equations like those represented by Eq. (6.102) for each region. In this case, it is necessary to solve the equations in each region and satisfy boundary conditions at every interface as well as at the reactor surface. This procedure can only be carried out in a practical way with a high-speed electronic computer, and many computer programs have been written for solving the multigroup equations.

For an infinite reactor, the fluxes are independent of position, ϕ_g = constant, $\nabla^2\phi_g = 0$, and the multigroup equations reduce to a set of linear algebraic equations with constant coefficients in the unknowns $\phi_1, \phi_2, \ldots, \phi_N$. Since these are homogeneous equations, according to Cramer's rule there can be no solutions other than $\phi_1 = \phi_2 = \cdots = \phi_N = 0$ unless the determinant of the coefficients of the fluxes is equal to zero, which is actually the requirement for the criticality of the infinite system. By changing the concentration of the fuel, the value of the determinant also changes. The concentration for which the determinant is zero is the composition required for the system to be critical. The *relative* magnitudes of the fluxes $\phi_1, \phi_2, \ldots$ can be found by solving the multigroup linear equations using values for the coefficients computed at the critical composition. *Absolute* values of the fluxes are not obtained, however, because these depend on the reactor power level. Nevertheless, the relative values of the flux are sufficient to estimate the neutron energy distribution or neutron spectrum in the reactor.

Such calculations of neutron spectra in infinite reactors are used in reactor design in the following way. Because of the complexity of the multigroup equations for a finite, multiregion reactor, it is often too costly in computer time to utilize a large number of groups in multigroup computations, especially in the early stages of the design of a reactor, when these computations must be carried out repeatedly with varying design parameters. It has been found more practical, therefore, to perform the initial design calculations for the finite reactor using only a few groups, but with multigroup constants computed using neutron spectra derived from multigroup calculations carried out with a large number of groups for the infinite reactor. Only when the design of a reactor has been more or less decided upon are large-scale, space-dependent multigroup calculations undertaken. These provide accurate estimates of the critical mass, the flux and power distributions, and other properties of the system.

6.8 HETEROGENEOUS REACTORS

Up to this point it has been assumed that the reactor core consists of a homogeneous mixture of fissile material, coolant, and, if the reactor is thermal, moderator, whereas in most reactors the fuel is actually contained in *fuel rods* of one sort or another. Such nonhomogeneous reactors are divided into two classes. If the neutron mean free path at *all* neutron energies is large compared with the thickness of the fuel rods, then it is unlikely that a neutron will have more than one successive collision within any one fuel rod. In this case, the reactor core is homogeneous as far as the neutrons are concerned and the reactor is called *quasi-homogeneous.* On the other hand, if the neutron mean free path at some energy is comparable to or smaller than the thickness of a fuel rod, then the neutrons may undergo several collisions within a fuel rod, the fuel and moderator must be treated as separate regions, and the reactor is said to be *heterogeneous.* Reactors with fuel rods in the form of thin plates are often of the quasi-homogeneous type, especially if the fuel is highly enriched and therefore present in the rod at low concentration. Most modern (thermal) power reactors, however, are fueled with only slightly enriched (a few weight percent) uranium, the fuel rods are not thin, and these reactors are heterogeneous.

Since from the standpoint of the neutrons a quasi-homogeneous reactor is in fact homogeneous, the formulas developed in the preceding sections for homogeneous systems can be used with quasi-homogeneous reactors. It is merely necessary in the computations to use the atom densities for the equivalent homogeneous reactor as shown in the following example.

Example 6.10 The fuel plates of a mobile thermal reactor are thin fuel "sandwiches" as shown in Fig. 6.6. The cladding is zirconium and the "meat" is a mixture of ^{235}U and zirconium, containing approximately 150 atoms of zirconium for

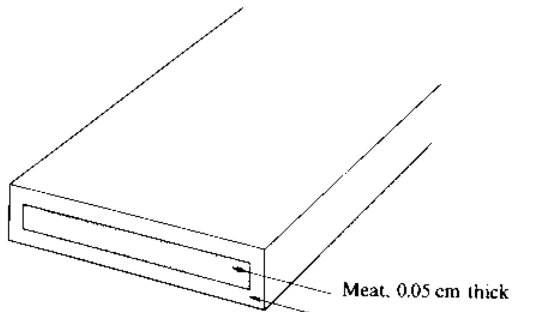

Fig. 6.6 A fuel plate sandwich.

every atom of uranium. Both the cladding and the meat are 0.05 cm thick. The plates are immersed in a water moderator-coolant in such a way that the total volumes of the plates and the water are the same. At room temperature: (a) What is the mean free path of thermal neutrons in the zirconium cladding and in the meat? (b) What is the value of k_∞ for the system?

Solution.

a) According to Table II.3, the atom density of zirconium, N_Z, in units of 10^{24}, is 0.0429, and its total cross section at thermal is approximately 6.6 b. The mean free path in the cladding is therefore

$$\lambda = 1/\Sigma_t = 1/0.0429 \times 6.6 = 3.5 \text{ cm. } [Ans.]$$

To find the mean free path in the meat it is necessary to compute Σ_t for the ^{235}U–zirconium mixture. Since the uranium is present essentially only as an impurity, the atom concentration of the zirconium in the meat is the same as in the cladding, that is, $N_Z = 0.0429$. The atom concentration of the ^{235}U is then $N_{25} = 0.0429/150 = 2.86 \times 10^{-4}$. For ^{235}U,

$$\sigma_t = \sigma_a + \sigma_s = 681 + 8.9 \simeq 690 \text{ b.}$$

Thus for the meat*

$$\Sigma_t = 2.86 \times 10^{-4} \times 690 + 0.0429 \times 6.6 = 0.480 \text{ cm}^{-1}.$$

The mean free path is then

$$\lambda = 1/\Sigma_t = 2.1 \text{ cm. } [Ans.]$$

*This calculation would appear to violate the provisions of Section 3.3 that thermal scattering cross sections must not be added. However, most of the contribution from the fuel to Σ_t is from absorption, and this can be added to scattering.

The mean free path in the cladding is 3.5/0.05 = 70 times larger than the thickness of the cladding, while it is 2.1/0.05 = 42 times larger in the meat. The reactor core is clearly quasi-homogeneous.

b) As usual, $k_\infty = \eta_T f$, where $\eta_T = 2.065$ for ^{235}U at room temperature. To compute f, the atom densities of water, zirconium, and ^{235}U of the equivalent homogenized mixture must be used. Since the water and zirconium each occupy one-half the volume, their respective atom densities are just one-half their normal values. In units of 10^{24}:

$$N_W = 0.5 \times 0.0334 = 0.0167,$$

where 0.0334 is the molecular density of unit-density water, and

$$N_Z = 0.5 \times 0.0429 = 0.0215.$$

The uranium is present in a volume equal to one-third of the sandwich or one-sixth of the total volume and its atom density there is 2.86×10^{-4} so that for the mixture

$$N_{25} = 2.86 \times 10^{-4}/6 = 4.77 \times 10^{-5}.$$

At 0.0253 eV, the microscopic absorption cross sections of ^{235}U, zirconium, and water (per molecule) are 681 b, 0.185 b, and 0.664 b, respectively. At 20°C, the non-$1/v$ factor for ^{235}U is 0.978. The value of f is then

$$f = \frac{4.77 \times 10^{-5} \times 681 \times 0.978}{4.77 \times 10^{-5} \times 681 \times 0.978 + 0.0215 \times 0.185 + 0.0167 \times 0.664} = 0.6783.$$

Thus,

$$k_\infty = 2.065 \times 0.6783 = 1.401. \ [Ans.]$$

Calculations for heterogeneous thermal reactors are considerably more difficult than those for homogeneous reactors. It is still possible to write $k_\infty = \eta_T f p \epsilon$, and to use the equations for criticality derived earlier, but the factors in these formulas must be computed with some care.

The Value of η_T

Most heterogeneous reactors are fueled with natural or partially enriched uranium in metallic or oxide form. The term "fuel" then refers to a mixture of ^{235}U, ^{238}U, and oxygen. In either case, η_T can be found from the formula

$$\eta_T = \frac{\nu_{25}\overline{\Sigma}_{f25}}{\overline{\Sigma}_{a25} + \overline{\Sigma}_{a28}}, \tag{6.103}$$

since the absorption cross section of oxygen is essentially zero.

Example 6.11 Compute the value of η_T for natural uranium at room temperature.

Solution. It is convenient first to rewrite Eq. (6.103) in the form

$$\eta_T = \frac{\nu_{25}\overline{\sigma}_{f25}}{\overline{\sigma}_{a25} + \dfrac{N_{28}}{N_{25}}\overline{\sigma}_{a28}}.$$

The ratio $N_{28}/N_{25} = 138$ for natural uranium. Then using the values $\nu_{25} = 2.42$, $\sigma_{f25} = 582$ b, g_{f25} (20°C) $= 0.976$, $\sigma_{a25} = 681$ b, g_{a28} (20°C) $= 0.978$, $\sigma_{a28} = 2.70$ b, g_{a28} (20°C) $= 1.0017$ gives

$$\eta_T = \frac{2.42 \times 582 \times 0.976}{681 \times 0.978 + 138 \times 2.70 \times 1.0017} = 1.32. \ [Ans.]$$

Thermal Utilization

The thermal utilization is defined as the probability that a thermal neutron, if it is ultimately absorbed in the core, is, in fact, absorbed in fuel, This, in turn, is equal to the ratio of the number of neutrons which are absorbed in fuel per second to the number absorbed per second in both fuel and moderator.*

The number of neutrons absorbed in the fuel per second is given by

$$\int_{V_F} \overline{\Sigma}_{aF}\phi_T \, dV,$$

where $\overline{\Sigma}_{aF}$ is the macroscopic thermal absorption cross section of the fuel and the integration is carried out over the volume of the fuel V_F. This number can also be written as

$$\overline{\Sigma}_{aF}\overline{\phi}_{TF} V_F,$$

where $\overline{\phi}_{TF}$ is the average thermal flux in the fuel. The number of neutrons absorbed per second in the moderator is given by a similar expression, so that f becomes

$$f = \frac{\overline{\Sigma}_{aF}\overline{\phi}_{TF} V_F}{\overline{\Sigma}_{aF}\overline{\phi}_{TF} V_F + \overline{\Sigma}_{aM}\overline{\phi}_{TM} V_M}. \qquad (6.104)$$

By dividing the numerator and denominator of Eq. (6.104) by $\overline{\phi}_{TF}$, f can be put in the form

$$f = \frac{\overline{\Sigma}_{aF} V_F}{\overline{\Sigma}_{aF} V_F + \overline{\Sigma}_{aM} V_M \zeta}, \qquad (6.105)$$

*It is assumed here that the reactor core consists of only fuel and moderator. In real reactors, cladding, structural materials, control rods, and other materials are always present.

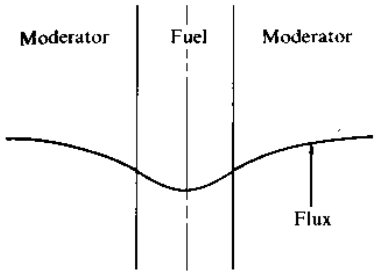

Fig. 6.7 Neutron flux in and near a fuel rod.

where

$$\zeta = \overline{\phi}_{TM}/\overline{\phi}_{TF} \tag{6.106}$$

is called, for reasons discussed later, the *thermal disadvantage factor.*

The values of ϕ_{TF} and ϕ_{TM}, cannot be calculated exactly except by numerical methods. However, the qualitative behavior of the flux in the fuel and moderator is shown in Fig. 6.7. It will be seen that the flux is smaller in the fuel than in the moderator. This is to be expected on physical grounds, since the absorption cross section of the fuel is so much higher than that of the moderator. It also follows that $\overline{\phi}_{TF}$ is smaller than $\overline{\phi}_{TM}$, so that from Eq. (6.106), ζ is greater than unity. This fact has important consequences that will be considered later in this section.

Although f cannot be calculated analytically, it is possible to derive approximate formulas for this parameter. In these approximations, an infinite array or *lattice* of fuel rods is divided into *unit cells* as shown in Fig. 6.8. Calculations of the rate at which neutrons are absorbed in the fuel and in the moderator are then carried out in the *equivalent cylindrical cell,* that is, the cell whose volume is equal to that of the noncylindrical cell. This procedure is called the *Wigner-Seitz method.*

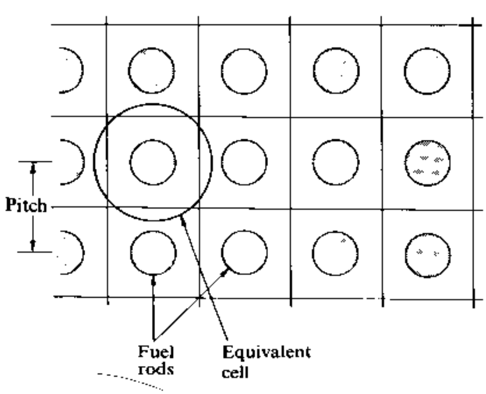

Fig. 6.8 A square lattice of fuel rods showing an equivalent cell.

TABLE 6.4
Thermal diffusion lengths of fuel

Fuel	L_F, cm
Natural uranium	1.55
U_3O_8	3.7

The results of these calculations are usually expressed in the following form:

$$\frac{1}{f} = \frac{\overline{\Sigma}_{aM} V_M}{\overline{\Sigma}_{aF} V_F} F + E, \tag{6.107}$$

where F and E are called *lattice functions*. In the simplest calculations of these functions, in which diffusion theory is assumed to be valid in both the fuel and the moderator, F and E are given by

$$F(x) = \frac{xI_0(x)}{2I_1(x)} \tag{6.108}$$

and

$$E(y, z) = \frac{z^2 - y^2}{2y}\left[\frac{I_0(y)K_1(z) + K_0(y)I_1(z)}{I_1(z)K_1(y) - K_1(z)I_1(y)}\right]. \tag{6.109}$$

In these formulas,

$$x = a/L_F, \qquad y = a/L_M, \qquad z = b/L_M,$$

where a is the radius of the fuel rod, b is the radius of the equivalent cell, and L_F and L_M are the thermal diffusion lengths of the fuel and moderator, respectively. A short list of values of L_F is given in Table 6.4. The functions I_n and K_n, where $n = 0$ and 1, are called *modified Bessel functions*. A short table of these functions is given in Appendix V.

Equations (6.108) and (6.109) are reasonably accurate for reactors in which $a \ll b$, that is, reactors in which the moderator volume is much larger than the fuel volume. This is usually the case in large gas cooled reactors. However, these equations provide only rough estimates of f for reactors with more tightly packed lattices. Most modern power reactors are of this latter type, and more complicated formulas must be used for these systems.

It may be mentioned that when x, y, and z are less than about 0.75, the following series expansions can be used with good accuracy to compute F and E for cylindrical rods:

$$F(x) = 1 + \frac{1}{2}\left(\frac{x}{2}\right)^2 - \frac{1}{12}\left(\frac{x}{2}\right)^4 + \frac{1}{48}\left(\frac{x}{2}\right)^6 - \cdots, \tag{6.110}$$

$$E(y, z) = 1 + \frac{z^2}{2}\left[\frac{z^2}{z^2 - y^2}\ln\left(\frac{z}{y}\right) - \frac{3}{4} + \frac{y^2}{4z^2} + \cdots\right]. \tag{6.111}$$

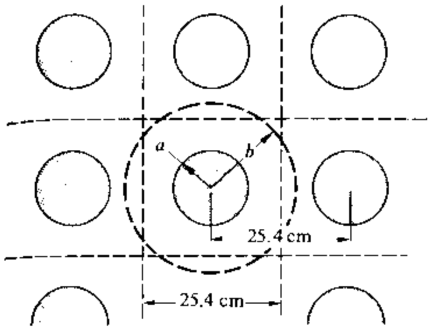

Fig. 6.9 A square lattice with 25.4-cm pitch.

Example 6.12 The core of an experimental reactor consists of a square lattice of natural uranium rods imbedded in graphite. The rods are 1.02 cm in radius and 25.4 cm apart.* Calculate the value of f for this reactor.

Solution. It is first necessary to determine the radius b of the cylindrical cell having the same volume as the unit cell. Both cells are shown in Fig. 6.9. These cells are of equal length, so for equal volumes it is necessary that

$$\pi b^2 = (25.4)^2,$$

and so

$$b = 25.4/\sqrt{\pi} = 14.3 \text{ cm}.$$

For natural uranium, $L_T = 1.55$ cm, while for graphite $L_T = 59$ cm. Thus

$$x = 1.02/1.55 = 0.658;$$
$$y = 1.02/59 = 0.0173;$$
$$z = 14.3/59 = 0.242.$$

Introducing these values into Eqs. (6.110) and (6.111) gives $F = 1.0532$ and $E = 1.0557$.

From Table II.3, at 0.0253 eV, $\Sigma_{aM} = 0.0002728 \text{ cm}^{-1}$ and $\Sigma_{aF} = 0.3668 \text{ cm}^{-1}$. Also, $V_M/V_F = (b^2 - a^2)/a^2 = 195.6$. Substituting into Eq. (6.107) then gives

$$\frac{1}{f} = \frac{0.0002728 \times 195.6}{0.3668} \times 1.0532 + 1.0557 = 1.2089,$$

so that

$$f = 0.8272. \text{ [\textit{Ans.}]}$$

---

*The distance between the centers of nearest rods, 25.4 cm in the example, is called the *lattice spacing* or *lattice pitch*.

Resonance Escape Probability

It will be recalled from Section 6.5 that the resonance escape probability, as the name implies, is the probability that a fission neutron will escape capture in resonances as it slows down to thermal energies. Analytical-numerical methods have been devised for computing p from nuclear cross section data, but these are beyond the scope of this text. However, many measurements of p have been carried out, and these have shown that p can be expressed approximately by the formula

$$p = \exp\left[-\frac{N_F V_F I}{\xi_M \Sigma_{sM} V_M}\right]. \tag{6.112}$$

Here N_F is the atom density of the fuel lump, in units of 10^{24}, V_F and V_M are the volumes of fuel and moderator, respectively, ξ_M is the average increase in lethargy per collision in the moderator (see Section 3.5), Σ_{sM} is the macroscopic scattering cross section of the moderator at resonance energies, and I is a parameter known as the *resonance integral*. Values of I for cylindrical fuel rods are well represented by the following empirical expression:

$$I = A + C/\sqrt{a\rho}, \tag{6.113}$$

in which A and C are measured constants given in Table 6.5, a is the fuel rod radius, and ρ is the density of the fuel. For convenience in making calculations, values of $\xi_M \Sigma_{sM}$ are given in Table 6.6.

It should be noted that according to Eq. (6.112), I has dimensions of cross section. The constants in Table 6.5, when introduced into Eq. (6.113), give I in barns. However, a must be given in cm, and ρ in g/cm^3. Also, it should be pointed out that although the values of the constants A and C in Table 6.5 are given for the pure resonance absorbers ^{238}U and ^{232}Th, these constants are also valid for slightly enriched uranium and for thorium containing a small quantity of converted ^{233}U. Small amounts of ^{235}U or ^{233}U in the fuel do not significantly alter the extent of resonance absorption in the ^{238}U or the ^{232}Th—the fuel is still essentially all ^{238}U or ^{232}Th, as the case may be.

TABLE 6.5
Constants for computing I*

Fuel	A	C
^{238}U (metal)	2.8	38.3
$^{238}UO_2$	3.0	39.6
^{232}Th (metal)	3.9	20.9
$^{232}ThO_2$	3.4	24.5

*Based on W.G. Pettus and M. N. Baldwin, Babcock and Wilcox Company reports BAW-1244, 1962 and BAW-1286, 1963.

TABLE 6.6
Values of $\xi_M \Sigma_{sM}$

Moderator	$\xi_M \Sigma_{sM}$
Water	1.46
Heavy water	0.178
Beryllium	0.155
Graphite	0.0608

Example 6.13 Calculate p for the lattice described in Example 6.12.

Solution. The rod radius is $a = 1.02$ cm, and the uranium density is 19.1 g/cm^3. From Eq. (6.113), the resonance integral is therefore

$$\begin{aligned} I &= 2.8 + 38.3/\sqrt{1.02 \times 19.1} \\ &= 11.5 \text{ b}. \end{aligned}$$

From Table II.3 $N_F = 0.0483$, in units of 10^{24}; from Table 6.6, $\xi_M \Sigma_{sM} = 0.0608$; and from Example 6.12 $V_M/V_F = 195.6$. Introducing these values into Eq. (6.112) gives

$$\begin{aligned} p &= \exp\left[-\frac{0.0483 \times 11.5}{0.0608 \times 195.6}\right] \\ &= \exp(-0.04670) = 0.9544. \ [\textit{Ans.}] \end{aligned}$$

Fast Fission

The fast fission factor, ϵ, the ratio of the total number of neutrons emitted from both thermal and fast fission to the number emitted in thermal fission alone, has been calculated and measured for a large number of heterogeneous reactors. Results of experiments with 1.50 cm diameter, slightly enriched uranium rods* in ordinary water are shown in Fig. 6.10. It will be observed that ϵ increases as the ratio of the uranium volume to the water increases. This would be expected physically. Thus with more uranium metal present in the system, the probability is greater that a fission neutron will strike a uranium nucleus before its energy falls below the fast fission threshold in a collision with the moderator. Also shown in Fig. 6.10 are computed values of ϵ for a homogeneous mixture of uranium and water. It will be evident from the figure that ϵ is not a sensitive function of the radius of the fuel rods.

*As noted earlier, most modern power reactors, at least those in the United States, use slightly (2–3%) enriched uranium. Since at low enrichments the ^{238}U content of the fuel is essentially independent of enrichment, the value of ϵ does not depend strongly on enrichment.

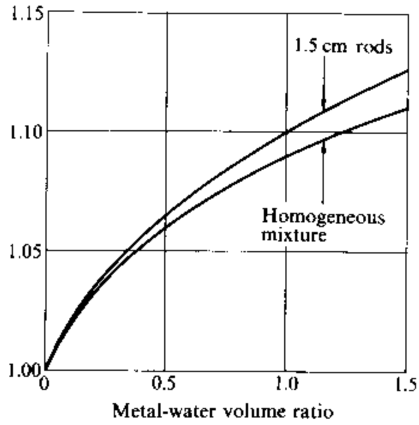

Fig. 6.10 The fast fission factor as a function of metal–water volume ratio.

Considerable additional data on ϵ, including data on moderators other than water, will be found in the report ANL-5800 (see References).

The Value of k_∞

It is of interest to compare the values of the parameters in the four-factor formula for a heterogeneous thermal reactor with those of a homogeneous mixture of the same materials. Consider first the thermal utilization, which for the heterogeneous system is given by Eq. (6.104) or (6.105):

$$f_{\text{hetero}} = \frac{\overline{\Sigma}_{aF}\overline{\phi}_{TF}V_F}{\overline{\Sigma}_{aF}\overline{\phi}_{TF}V_F + \overline{\Sigma}_{aM}\overline{\phi}_{TM}V_M} = \frac{\overline{\Sigma}_{aF}V_F}{\overline{\Sigma}_{aF}V_F + \overline{\Sigma}_{aM}V_M\zeta},$$

where ζ is the disadvantage factor (cf. Eq. 6.106). If the system is homogenized, the terms $\overline{\Sigma}_{aF}\overline{\phi}_{TF}V_F$ and $\overline{\Sigma}_{aM}\overline{\phi}_{TM}V_M$ give the total absorption in the fuel and moderator. However, in a homogeneous system the fluxes in the fuel and moderator are the same, that is, $\overline{\phi}_{TF} = \overline{\phi}_{TM}$, so that $\zeta = 1$, and therefore

$$f_{\text{homo}} = \frac{\overline{\Sigma}_{aF}V_F}{\overline{\Sigma}_{aF}V_F + \overline{\Sigma}_{aM}V_M}. \tag{6.114}$$

It was shown earlier that ζ is greater than unity for a heterogeneous reactor. From a comparison of Eqs. (6.105) and (6.114) it follows that

$$f_{\text{hetero}} < f_{\text{homo}}. \tag{6.115}$$

The physical reason why f is smaller for the heterogeneous system than for its homogenized equivalent is that in the heterogeneous case, as noted earlier, the flux is lower in the fuel than it is in the moderator. This depression in the flux is caused by the fact that some of the neutrons entering the fuel from the moderator are absorbed near the surface of the fuel, and they simply do not survive to contribute to the flux in its interior. The outer portion of the fuel thus shields the interior, and this state of affairs is known as *self-shielding*.

What is a disadvantage for the thermal utilization, however, is necessarily an advantage as far as the resonance escape probability is concerned. Thus, while self-shielding decreases the absorption of thermal neutrons in the fuel, it also reduces the number of neutrons which are absorbed parasitically by resonance absorbers in the fuel. It follows, therefore, that the resonance escape probability for a heterogeneous system is *larger* than for the equivalent homogeneous mixture. Furthermore, since the average absorption cross section of the fuel at resonance energies is much larger than the absorption cross section at thermal energies, the depression in the flux of resonance neutrons is greater than it is for thermal neutrons. As a consequence, the decrease in f for the heterogeneous system is more than offset by the increase in p.

The fast fission factor for a heterogeneous reactor is also larger than for the homogeneous system, as may be seen in Fig. 6.10. This is due to the fact that in the heterogeneous case the fission neutrons pass through a region of pure fuel, where they are likely to induce fast fission, before they encounter the moderator. The effect of heterogeneity is not as pronounced with ϵ, however, as it is with f and p.

In view of the fact that

$$(fp)_{\text{hetero}} > (fp)_{\text{homo}}$$

and

$$\epsilon_{\text{hetero}} > \epsilon_{\text{homo}},$$

it follows that k_∞ for a heterogeneous reactor is greater than for its homogeneous equivalent.* This result is of considerable importance in the design of certain reactors. In particular, the maximum value of k_∞ for a homogeneous mixture of natural uranium and graphite is only 0.85 and such a reactor cannot become critical. By lumping the fuel into a heterogeneous lattice, k_∞ can be made sufficiently greater than unity that a critical reactor can be constructed using these materials.

REFERENCES

General

Glasstone, S., and M. C. Edlund, *The Elements of Nuclear Reactor Theory*. Princeton, N.J.: Van Nostrand, 1952. Chapters 7–9.

Glasstone, S., and A. Sesonske, *Nuclear Reactor Engineering*, 3rd ed., New York: Van Nostrand, 1981, Chapter 4.

Isbin, H. S., *Introductory Nuclear Reactor Theory*. New York: Reinhold, 1963, Chapters 6–8.

*This conclusion is valid for heterogeneous reactors fueled with uranium enriched to about 5% in ^{235}U. Above this enrichment, p is of less overall importance in determining the value of k_∞, and lumping the fuel tends only to reduce the overall absorption of neutrons by the fissile material.

Lamarsh, J. R., *Introduction to Nuclear Reactor Theory.* Reading, Mass.: Addison-Wesley, 1966, Chapters 6–11.

Liverhant, S. E., *Elementary Introduction to Nuclear Reactor Physics.* New York: Wiley, 1960.

Meem, J. L., *Two Group Reactor Theory.* New York: Gordon and Breach, 1964, Chapters 3, 7, and 9.

Murray, R. L., *Nuclear Reactor Physics.* Englewood Cliffs, N.J.: Prentice-Hall, 1957, Chapters 3–5.

Sesonske, A., *Nuclear Power Plant Analysis.* U.S. Atomic Energy Commission report TID-26241, 1973, Chapter 5.

Zweifel, P. F., *Reactor Physics.* New York: McGraw-Hill, 1973, Chapter 3.

Computer Codes

The Argonne Code Center at the Argonne National Laboratory, Argonne, Illinois, collects, maintains, and distributes a library of computer programs or codes for the solution of problems in nuclear reactor design and engineering. The Center also provides the documentation necessary to understand and use the various codes.

Reactor Data

Considerable useful data will be found in the Argonne National Laboratory report *Reactor Physics Constants,* ANL-5800, L. J. Templin, Editor, 1963.

Bessel Functions

These functions are tabulated in many places including:

Abramowitz, M., and I. A. Stegun, Editors, *Handbook of Mathematical Tables.* Washington, D.C.: National Bureau of Standards, 1964.

Etherington, H., Editor, *Nuclear Engineering Handbook.* New York: McGraw-Hill, 1958.

PROBLEMS

Note: Make all computations at room temperature, unless otherwise stated, and take the recoverable energy per fission to be 200 MeV.

6.1 Calculate the fuel utilization and infinite multiplication factor for a fast reactor consisting of a mixture of liquid sodium and plutonium, in which the plutonium is present to 3.0 w/o. The density of the mixture is approximately 1.0 g/cm^3.

6.2 The core of a certain fast reactor consists of an array of uranium fuel elements immersed in liquid sodium. The uranium is enriched to 25.6 w/o in ^{235}U and comprises 37 percent of the core volume. Calculate for this core (a) the average atom densities of sodium, ^{235}U, and ^{238}U; (b) the fuel utilization; (c) the value of η; (d) the infinite multiplication factor.

6.3 A bare cylindrical reactor of height 100 cm and diameter 100 cm is operating at a steady-state power of 20 MW. If the origin is taken at the center of the reactor, what is the power density at the point $r = 7$ cm, $z = -22.7$ cm?

6.4 In a spherical reactor of radius 45 cm the fission rate density is measured as 2.5×10^{11} fissions/cm^3-sec at a point 35 cm from the center of the reactor. (a) At what steady-state power is the reactor operating? (b) What is the fission rate density at the center of the reactor?

6.5 Using the flux function given in Table 6.2 for a critical finite cylindrical reactor, derive the value of the constant A.

6.6 The core of a certain reflected reactor consists of a cylinder 10 ft high and 10 ft in diameter. The measured maximum-to-average flux is 1.5. When the reactor is operated at a power level of 825 MW, what is the maximum power density in the reactor in kW/liter?

6.7 Suppose the reactor described in Example 6.3 were operated at a thermal power level of 1 kilowatt. How many neutrons would escape from the reactor per second? [*Hint:* See Example 6.4.]

6.8 Show that in a one-group model the power produced by a reactor per unit mass of fissile material is given by

$$\frac{\text{watts}}{\text{g}} = \frac{\text{kW}}{\text{kg}} = \frac{3.2 \times 10^{-11} \sigma_f \bar{\phi} N_A}{M_F},$$

where σ_f is the one-group fission cross section, $\bar{\phi}$ is the average one-group flux, N_A is Avogadro's number, and M_F is the gram atomic weight of the fuel.

6.9 (a) Estimate the critical radius of a hypothetical bare spherical reactor having the same composition as the reactor in Problem 6.1. (b) If the reactor operates at a thermal power level of 500 MW, what is the maximum value of the flux? (c) What is the probability that a fission neutron will escape from the reactor?

6.10 An infinite slab of moderator of thickness $2a$ contains at its center a thin sheet of ^{235}U of thickness t. Show that in one-group theory the condition for criticality of this system can be written approximately as

$$\frac{2D}{(\eta - 1)L\Sigma_{aF}t} \coth \frac{a}{L} = 1,$$

where D and L are the diffusion parameters of the moderator and Σ_{aF} is the macroscopic absorption cross section of the ^{235}U.

6.11 A large research reactor consists of a cubical array of natural uranium rods in a graphite moderator. The reactor is 25 ft on a side and operates at a power of 20 MW. The average value of $\bar{\Sigma}_f$ is 2.5×10^{-3} cm^{-1}. (a) Calculate the buckling. (b) What is the maximum value of the thermal flux? (c) What is the average value of the thermal flux? (d) At what rate is ^{235}U being consumed in the reactor?

6.12 Show that with a recoverable energy per fission of 200 MeV, the power of a ^{235}U fueled reactor operating at the temperature T is given by either of the following expressions:

$$P = 4.73\, m_F g_F(T) \bar{\phi}_0 \times 10^{-14}\ \text{MW}$$

or

$$P = 7.19\, m_F g_F(T) T^{-1/2} \bar{\phi}_T \times 10^{-13}\ \text{MW},$$

where m_F is the total amount in kg of ^{235}U in the reactor, $g_F(T)$ is the non-$1/v$ factor for fission, $\bar{\phi}_0$ is the average 2200 meters-per-second flux, and $\bar{\phi}_T$ is the average thermal flux.

6.13 Solve the following equations:

$$\begin{aligned} 3x + 4y + 7z &= 16, \\ x - 6y + z &= 2, \\ 2x + 3y + 3z &= 12. \end{aligned}$$

6.14 Solve the following equations:

$$\begin{aligned} 3.1x + 4.0y + 7.2z &= 0, \\ x - 5.0y - 9.0z &= 0, \\ 7x + 4.5y + 8.1z &= 0. \end{aligned}$$

6.15 A homogeneous solution of ^{235}U and H_2O contains 10 grams of ^{235}U per liter of solution. Compute (a) the the atom density of ^{235}U and the molecular density of H_2O; (b) the thermal utilization; (c) the thermal diffusion area and length; (d) the infinite multiplication factor.

6.16 Compute the thermal diffusion length for homogeneous mixtures of ^{235}U and the following moderators at the given fuel concentrations and temperatures.
(a) Graphite: $N(25)/N(C) = 4.7 \times 10^{-6}$; $T = 200°C$.
(b) Beryllium: $N(25)/N(\text{Be}) = 1.3 \times 10^{-5}$; $T = 100°C$.
(c) D_2O: $N(25)/N(D_2O) = 1.4 \times 10^{-6}$; $T = 20°C$.
(d) H_2O: $N(25)/N(H_2O) = 9.2 \times 10^{-4}$; $T = 20°C$.

6.17 Consider a critical bare slab reactor 200 cm thick consisting of a homogeneous mixture of ^{235}U and graphite. The maximum thermal flux is 5×10^{12} neutrons/cm^2-sec. Using modified one-group theory, calculate: (a) the buckling of the reactor; (b) the critical atomic concentration of uranium; (c) the thermal diffusion area; (d) the value of k_∞; (e) the thermal flux and current throughout the slab; (f) the thermal power produced per cm^2 of this slab.

6.18 The binding energy of the last neutron in ^{13}C is 4.95 MeV. Estimate the recoverable energy per fission in a large graphite moderated, ^{235}U fueled reactor from which there is little or no leakage of neutrons or γ-rays.

6.19 Calculate the concentrations in grams per liter of (1) ^{235}U, (2) ^{233}U, and (3) ^{239}Pu required for criticality of infinite homogeneous mixtures of these fuels and the following moderators: (a) H_2O, (b) D_2O, (c) Be, (d) graphite.

6.20 A bare spherical reactor 50 cm in radius is composed of a homogeneous mixture of ^{235}U and beryllium. The reactor operates at a power level of 50 thermal kilowatts. Using modified one-group theory, compute: (a) the critical mass of ^{235}U; (b) the thermal flux throughout the reactor; (c) the leakage of neutrons from the reactor; (d) the rate of consumption of ^{235}U.

6.21 It is proposed to store H_2O solutions of fully enriched uranyl (ū'rȧ-nil) sulfate ($^{235}UO_2SO_4$) with a concentration of 30 g of this chemical per liter. Is this a safe procedure when using a tank of unspecified size?

6.22 A bare thermal reactor in the shape of a cube consists of a homogeneous mixture of ^{235}U and graphite. The ratio of atom densities is $N_F/N_M = 1.0 \times 10^{-5}$ and the fuel temperature is 250°C. Using modified one-group theory, calculate: (a) the critical dimensions; (b) the critical mass; (c) the maximum thermal flux when the reactor operates at a power of 1 kW.

6.23 The original version of the Brookhaven Research Reactor consisted of a cube of graphite which contained a regular array of natural uranium rods, each of which was located in an air channel through the graphite. When the reactor was operated at a thermal power level of 22 MW, the average fuel temperature was approximately 300°C and the maximum thermal flux was 5×10^{12} neutrons/cm^2-sec. The average values of L_T^2 and τ_T were 325 cm^2 and 396 cm^2, respectively, and $k_\infty = 1.0735$. (a) Calculate the critical dimensions of the reactor. (b) What was the total amount of natural uranium in the reactor?

6.24 Using one-group theory, derive expressions for the flux and the condition for criticality for the following reactors: (a) an infinite slab of thickness a, infinite reflector on both sides; (b) an infinite slab of thickness a, reflectors of thickness b on both sides; (c) a sphere of radius R, reflector of thickness b.

6.25 The core of a spherical reactor consists of a homogeneous mixture of ^{235}U and graphite with a fuel–moderator atom ratio $N_F/N_M = 6.8 \times 10^{-6}$. The core is surrounded by an infinite graphite reflector. The reactor operates at a thermal power of 100 kW. Calculate: (a) the value of k_∞; (b) the critical core radius; (c) the critical mass; (d) the reflector savings; (e) the thermal flux throughout the reactor; (f) the maximum-to-average flux ratio.

6.26 Estimate the new critical radius and the critical mass of a reactor of the same composition as described in Problem 6.20 when the core is surrounded by an infinite beryllium reflector.

6.27 An infinite cylindrical reactor core of radius R is surrounded by an infinitely thick reflector. Using one-group theory, (a) find expressions for the fluxes in the core and reflector; (b) show that the condition for criticality is

$$D_c B \frac{J_1(BR)}{J_0(BR)} = \frac{D_r}{L_r} \frac{K_1(R/L_r)}{K_0(R/L_r)},$$

where J_0 and J_1 are ordinary Bessel functions and K_0 and K_1 are modified Bessel functions.

6.28 The core of an infinite planar thermal reactor consists of a solution of ^{239}Pu and H_2O with a plutonium concentration of 8.5 g/liter. The core is reflected on both faces by infinitely thick H_2O reflectors. Calculate: (a) the reflector savings, (b) the critical thickness of the core, (c) the critical mass in g/cm^2.

6.29 The core of a thermal reactor consists of a sphere, 50 cm in radius, which contains a homogeneous mixture of ^{235}U and ordinary H_2O. This core is surrounded by an infinite H_2O reflector. (a) What is the reflector savings? (b) What is the critical mass? (c) If the maximum thermal flux is 1×10^{13} neutrons/cm^2-sec, at what power is the reactor operating? [*Hint:* Compute M_T^2 for core assuming reactor is bare and of radius 50 cm. Use this value of M_T^2 to estimate δ and then compute new M_T^2 assuming reactor is bare and of radius $50 + \delta$. Iterate until convergence is obtained.]

6.30 A spherical breeder reactor consists of a core of radius R surrounded by a breeding blanket of thickness b. The infinite multiplication factor for the core $k_{\infty c}$ is greater than unity while that for the blanket $k_{\infty b}$ is less than unity. Using one-group theory derive the critical conditions and expressions for the flux throughout the reactor.

6.31 Plate-type fuel elements for an experimental reactor consist of sandwiches of uranium and aluminum. Each sandwich is 7.25 cm wide and 0.16 cm thick. The cladding is aluminum which is 0.050 cm thick. The meat is an alloy of fully enriched uranium and aluminum which has 20 % uranium and has a density of approximately 3.4 g/cm^3. (a) Estimate the mean free path of thermal neutrons in the meat and cladding in this fuel element. (b) Is a reactor fueled with these elements quasi-homogeneous or heterogeneous?

6.32 The fuel sandwiches in the reactor in the preceding problem are braised into aluminum holders and placed in a uniform array in an ordinary water moderator. The (total) metal–water volume ratio is 0.73, and there are in total 120 atoms of aluminum per atom of uranium. Calculate (a) the thermal utilization; (b) k_∞; (c) the thermal diffusion length. [*Note:* In part (c), compute the value of $\bar{D}$ for the homogeneous mixture by adding together the macroscopic transport cross sections of the aluminum and water. For Σ_{tr} of water use $1/3\bar{D}$, where $\bar{D}$ is the experimental value of $\bar{D}$ for water, corrected of course for density.]

6.33 A heterogeneous uranium–water lattice consists of a square array of natural uranium rods 1.50 cm in diameter with a pitch of 2.80 cm. Calculate (a) the radius of the equivalent cell; (b) the uranium–water volume ratio V_F/V_M; (c) the thermal utilization; (d) the resonance escape probability; (e) the fast fission factor (from Fig. 6.10); (f) k_∞.

6.34 Repeat the calculations of Problem 6.33 for the case in which the uranium is enriched to 2.5 $^w\!/_o$ in ^{235}U.

6.35 In a hexagonal lattice (also called a triangular lattice), each fuel rod is surrounded by six nearest neighboring rods, equally spaced at a distance equal to the pitch s of the lattice. Show that the radius of the equivalent cell in this lattice is given by

$$b = 0.525s.$$

6.36 Calculate k_∞ for a hexagonal lattice of 1.4-cm-radius natural uranium rods and graphite, if the lattice pitch is 20 cm. [*Note:* The fast fission factor for this lattice is 1.03.]

INDEX

Note: Numbers in parentheses refer to problems.